Validating Abstract Representations of Spatial Population Data while considering Disclosure Avoidance

by

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Abstract

This paper furthers a research agenda for modeling populations along spatial networks and expands upon an empirical analysis to a full U.S. county (Gaboardi, 2019, Ch. 1,2). Specific foci are the necessity of, and methods for, validating and benchmarking spatial data when conducting social science research with aggregated and ambiguous population representations. In order to promote the validation of publicly-available data, access to highly-restricted census microdata was requested, and granted, in order to determine the levels of accuracy and error associated with a network-based population modeling framework. Primary findings reinforce the utility of a novel network allocation method—populated polygons to networks (pp2n) in terms of accuracy, computational complexity, and real runtime (Gaboardi, 2019, Ch. 2). Also, a pseudo-benchmark dataset's performance against the true census microdata shows promise in modeling populations along networks.

Keyword*:* network allocation, Master Address File, population representation

[*] This CES working paper is based on Chapter 3 of Populated Polygons to Networks: A Population-Centric Approach to Spatial Network Allocation (Gaboardi, 2019).

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1 Introduction

A large portion of social science research is conducted utilizing individual census response data that are aggregated into a hierarchy of census geographies. But how well do these aggregated, publicly-published data (and secondary datasets produced with that data), actually represent the reality of population distributions and how can researchers confirm their robustness? Moreover, social scientists are interested in anthropocentric phenomena, frequently at scales larger than a single neighborhood, and human activity is generally constrained to spatial networks (roads, etc.). So determining whether the results of research involving networks based on varying methods of network allocation can be verified against the elusive "truth" has inherent value. Here the aim is to determine whether the $pp2n$ method for spatial network allocation performs similarly within a larger spatial extent when comparing the segment-based population measures introduced in Gaboardi (2019, Ch. 2). This will be accomplished through repeating the analyses at the county level that are performed within a single tract and testing against a known-truth dataset, the Master Address File (MAF), in order to determine the validity of representing population abstractions along networks.

The validity of the aggregated data is assumed to be performed internally by the U.S. Census Bureau prior to release, along with implementing measures for protecting privacy, which will be discussed later in this paper. Validating collected and compiled data can be partitioned into two categories: attribute data and spatial data, where the former are variables such as *Age* and *Income* and the latter are concerned specifically with locational attributes. Variables, such as the geographic identifier (GEOID or FIPS — Federal Information Processing Series — codes), may be considered either attribute data or spatial data due to being inherently tied to the bounds of a specific census geography (state, county, tract, etc.).

Within the context of this paper the focus is specifically on understanding the uncertainty created in spatial data as it is aggregated for reasons of privacy and highlighting methods for determining the validity of data. Further, the aim is to provide confirmation of the soundness of publicly-published data, and derived products, for network-based population modeling against the known truth. As mentioned above, the underlying measure of validity for the results presented in the paper is based on the MAF, a nationwide file of geocoded locations that can be considered the "truth." The MAF is a database for all known living quarters in the U.S. and Puerto Rico with a (nearly) complete record of all residential addresses at a housing unit-level spatial resolution (Young et al., 2017). Therefore, having access to this dataset provides a crucial building block for the determination of data and results validity. Understandably, it is highly-restricted and inaccessible without a security clearance with the U.S. Census Bureau, known as Special Sworn Status.

The MAF was first developed in the 1990s from primarily the United States Postal Service's Delivery Sequence File, and implemented for the 2000 Decennial Census (Uhl, 2011). It is continuously updated as information becomes available, such as new addresses, updated information regarding old addresses, updated attributes, and improved precision of associated longitude and latitude. Following the initial implementation of the MAF for the 2000 Decennial Census it was merged with the publicly-available Topologically Integrated Geographic Encoding and Referencing (TIGER) database that is now known as the MAF/TIGER, which is the source of the recognizable

MTFCC (MAF/TIGER Feature Class Code) attribute frequently seen in census data products. As noted above, access to the MAF is highly restricted and governed by stringent policies concerning disclosure and privacy.

Before the methods of processing the MAF data are discussed, the concept of uncertainty in spatial data is discussed in more general terms prior to a narrow focus on uncertainty in census data and methods for validating spatially-centric census data. Context is provided for the publicly-available data and the daysmetricly-derived data of the complete study area of Leon County, FL from which the data in the empirical example of Gaboardi (2019, Ch. 2) was subset followed by a critique of the preprocessing dasymetric technique developed by Strode et al. (2018). After this critique, the primary research activity is resumed: an investigation and comparison of network segment population values derived from the methods introduced in Gaboardi (2019, Ch. 2). This is done by evaluating descriptive statistics for all network allocation scenarios, derived from both publicly-published census data and a derived product of internal census microdata. Finally, an evaluation of network-based distance versus Euclidean distance is performed for both unweighted distance and population-weighted distance. The sum goal of these scrutinizing reviews and analyses is to push forward a rigorous, multifaceted campaign of validation through an understanding of inherent spatial uncertainty, reasonable data review, critiquing quantitive methods, and developing a qualitative awareness of the study area. The core research topics here are validating the network allocation methods in this paper in general and specifically for measuring the accuracy of the $pp2n$ method against true population data. A secondary, though extremely valuable, research question is determining how the pseudo-benchmark dataset compares to the MAF-derived data.

2 Data Validation

2.1 Uncertainty in Spatial Data

Uncertainty, inaccuracy, and error are intertwined and, to a certain extent, generally considered unavoidable when working with spatial data, specifically spatial data within the social sciences. Therefore, the need to (attempt to) understand this mélange of ambiguity cannot be overstated when conducting research that has the potential to affect human lives. Uncertainty within a GIScience context can have various definitions and be the result of many underlying factors. According to Maffini et al. (1989) the issue of error with Geographic Information Systems (GIS) data can be broken down into two categories: (1) the types of errors and their causes, and (2) methods for handling said errors in derived data products. Category 1, error types and causes, can be further enumerated as (a) the features of the natural world being inherently continuous with few truly finite borders, (b) the limited capacity and precision of measurement and data collection with even the most sophisticated equipment, and (c) modeling selection, data generalization, and the bias involved with the communication of results. Category 2, solutions for dealing with error, encompasses (a) understanding the purpose for which the data will be used, including the scale, scope, and impact of associated error, and (b) using GIS and ancillary data to mediate the effects of error.

As stated in (2b) of the issues raised by Maffini et al. (1989), validating data through spatial and aspatial methods is crucial. Laurini and Thompson (1992) provide many examples of the need for validation within the context of spatial information systems, an alternative term for GIS. Particular issues pertaining directly to this line of research include data quality and precision (Laurini and Thompson, 1992, Ch. 3), the need for external validity (verifying the relationship between reality and data)(Laurini and Thompson, 1992, Ch. 5), geometric and spatial data transformations (Laurini and Thompson, 1992, Ch. 7), and aggregation (Laurini and Thompson, 1992, Ch. 12), which forms a central issue in this research along with its inverse: disaggregation. Because of the issues with representing spatial data stated in the previous section due dilligence must me performed, though is may be infeasible to verify the accuracy of every feature and attribute. Furthermore, uncertainty may be exacerbated because there may be no clear consensus on what actually constitutes geographic features (Mark et al., 2001).

Due to the varying nature of uncertainty in spatial data, there are many solutions to the many problems. Gabrosek and Cressie (2002) recognize how errors in locational data can drive errors in the associated attributes for data points. In order to solve this issue, a kriging method was developed for interpolating points in space more accurately. Similar to a main theme that is discussed in Gaboardi (2019, Ch. 4), Aerts et al. (2003) demonstrate how uncertainty and error in spatial data utilized as optimization problem parameters can propagate through a model and have a tangible impact on outcomes. Amatulli et al. (2007) allude to uncertainty in transforming points to polygons—transformed representation problem (TRP) case 3 from Fig. 2.1 in Gaboardi (2019, Ch. 2)—in interpolating the probability of wildfire ignition points in an area when only specific longitude–latitude coordinates are known through historical records. Though many issues relating to uncertainty in spatial data involve digital precision errors and sloppy record-keeping by experts, this is becoming more frequently not the sole cause.

Recent years have seen the proliferation of spatial data into the daily lives of the layman, not only those whose professions revolve around it like GIS technicians and surveyors. As more and more of the general public are exposed to spatial data, the need for vigilance in validating sources of this voluntarily contributed data becomes increasingly important (Flanagin and Metzger, 2008). Goodchild (2007) coined the term *volunteered geographic information* (VGI), which has since become a heavily-studied topic, while Haklay (2010) further evaluates the accuracy of a specific form of VGI: OpenStreetMaps (OpenStreetMap Contributors, 2017). In terms of assuring the quality of VGI, Goodchild and Li (2012) discuss the overarching issues and suggest approaches to ensure data robustness. The main themes surrounding the use of VGI from a scholarly research standpoint are spatial data quality (uncertainty) and the protection of contributor's privacy, both of which are extremely relevant to this line of research.

Although uncertainty quantification involving spatial data has a long history, it is still receiving considerable attention, and most likely will for many decades to come. Bielecka and Burek (2019) seem to suggest this while providing a longitudinal survey of spatial data quality and uncertainty in scholary research across academic subjects. Perhaps unsurprisingly, they find that the field of geography is one of the top ranking disciplines.

2.2 Validation within the (Spatial) Social Sciences

Validation is paramount for rigorous scientific research. As previously stated, however, much social science research relies on publicly-available data for testing. Specifically in regards to U.S. Census Bureau data, this means the least spatially aggregated data are population tabulations at the block level, the base unit of census geographies (U.S. Census Bureau, 2010b). This aggregation is essential for protecting respondents' privacy, but it also means that there is always a caveat when performing analyses based on aggregated, publicly-available data, even at the highest level of spatial resolution.

A significant amount of research[4] has been conducted in the past decade on the uncertainty of census data, specifically the decennial census and American Community Survey (ACS), with a particular focus on gaining a better understanding of the concept of a neighborhood. Some of these studies use only publicly-available data or only restricted-access census microdata, while others use a combination of the two. Spielman and Logan (2013) develop an approach for neighborhood identification using address-level census microdata from the late 19th century. Due to its historical nature (public microdata access is embargoed for 72 years), 100% of the processed microdata was freely available and four cities in the U.S. were selected as a subset. Spielman and Logan (2013) conclude that valuable and interesting insights are available through the utilization of microdata that are otherwise lost in the process of aggregation. Through the modeling results produced using synthetic data, Spielman et al. (2013) conclude that the spatial definition of a neighborhood plays a role in determining the extent to which said neighborhood affects health and behavioral characteristics of individuals who live in them. In the future, an extension of this study incorporating geocoded MAF data for validation in comparison with synthetic data would likely produce interesting results. In contrast to *ecological fallacy*, the inability to correctly infer individual characteristics from the statistical measures of aggregated units (Robinson, 1950), Fowler et al. (2019) introduce the concept of *contextual fallacy*, and a measure referred to as the *standard deviation of individual context* to elucidate the issues surrounding the assumption that an individual's context is equal to the context of the census geography in which they reside. To accomplish this, all 308,745,538 individual responses to the 2010 Decennial Census were used and matched to addresses of residence with an associated longitude and latitude through the Master Address File (MAF). Through this approach, Fowler et al. (2019) provide a deeper realization into measuring the uncertainty pertaining to spatial aggregation and individual context that would not have been possible without access to microdata. Prior to studying uncertainty in neighborhood context with individual household-level microdata, Spielman and Singleton (2015) investigated neighborhoods and uncertainty through the ACS. In their paper, Spielman and Singleton (2015) identify large margins of error for single variables in smaller areas that are particularly troublesome. Their solution is to combine multiple variables, instead of a single variable, to develop a measure for properly understanding the context of neighborhoods utilizing ACS data.

As noted above, the ACS is a valuable resource for researchers. However, in terms of uncertainty the ACS is far from perfect and currently much less reliable than the traditional form of the decennial

[4]Much of the research discussed in this section was the product of the NSF-Census Bureau Research Network (NCRN) and had played an active role in not only improving the utility of census data products, but also helping to ensure the continuance of federally-funded spatial and statistical research programs (Weinberg et al., 2018).

census. Spielman et al. (2014) describe the large margins of error associated the ACS in certain areas and the dangers of utilizing the ACS in research without clearly considering the implications of such large margins of error. Spielman and Folch (2015) revisit the issue of large margins of error in the ACS through regionalization, a spatial optimization method for building regions. Via regionalization, Spielman and Folch (2015) show that margins of error can be reduced through the creation of newly-synthesized census geographies. Bradley et al. (2017) also use regionalization to investigate spatial aggregation errors in which they advance a novel *criterion for spatial aggregation error* (CAGE). The ACS is utilized in an empirical example in Bradley et al. (2017) where it is shown to be effective at finding optimal regionalization solutions. Folch et al. (2016) find that the levels of uncertainty in the ACS are not uniform across the United States. Lower quality estimates (more uncertainty) appear to be associated more often with census tracts across the southern portion of the U.S. and vice versa across the northern U.S., concentrated in the Midwest.

In terms of the uncertainty in the decennial census, Logan et al. (2014) developed a database for mitigating error over time. Since the boundaries for census geographies have the potential to be redrawn prior to each decennial census it can never be assumed that a census geography has the same boundary (or even exists) from census to census. While the purpose is to provide proportional governmental representation, it creates issues for conducting longitudinal census research. The Longitudinal Tract Database (LTDB) developed by Logan et al. (2014) makes it possible to interpolate census tract estimates across decennial censuses from 1970 to 2010.

Randon-Furling et al. (2018) propose a unique approach for examining spatial dissimilarities and provide empirical examples for statistical geographies in Paris, France. They are building neighborhoods through a multi-focal perpective by focusing in at the micro-scale and determing the context of the area while pulling out to the metropolitan level. It would be very intersting to adapt the method to U.S. census geographic units with the potential to compare the results against the restricted-access MAF. Also, there exists the possibility of further validation of results through the integration of remote sensing techniques for neighborhood detection (Qi et al., 2019).

To succinctly synopsize the above research, the idea of defining a neighborhood for the purpose of scholarly research is frequently undercut by uncertainty due to the lack of (access to) high-resolution geographic data. But how can this core issue of uncertainty due to large aggregations of data and margin of error be resolved? The answer surely is not making available all census microdata, though there is potential in derived data products through the use of methods such as *populated polygons to networks* (*pp*2*n*), as proposed in Gaboardi (2019, Ch. 2). Not all the research discussed above dealt specifically with census microdata. However, all of the research was based on spatial techniques for measuring and correcting problems found in publicly-available census data. Many of these problems are caused by the various schema of data aggregation adopted by the U.S. Census Bureau for implementing a high standard of privacy. It is the legal responsibility of the U.S. Census Bureau to protect the privacy of respondents and confidentiality of respondent data against disclosures, accidental or otherwise. These laws were formally enacted with the adoption of Title 13 of the U.S. Code and will be discussed in the following section.

2.3 Disclosure and Privacy

The cornerstone of a census process is maintaining strict standards of confidentiality to keep the public's faith that their privacy will be protected while also providing high-quality data products. If the public should have any reason to doubt the full protection of their privacy, there will be a higher likelihood of falsified, incomplete, and misleading responses. Without accurate responses from the public, a census is virtually worthless. Therefore, the keystone of a census process is demonstrably protecting confidential data and respondent privacy. To this end, Title 13 of the U.S. Code was enacted in 1954, specifically Section 5, which protects the privacy of individual respondents, as well as businesses (U.S. Congress, 1954). An overview of U.S.C. Title 13, Section 5 taken from U.S. Census Bureau (2019c) can be found in Appendix B. For a chronological breakdown of the events that lead to the creation of Title 13 see U.S. Census Bureau (2002), and for a complete history of the issues of confidentiality and privacy relating the U.S. Census Bureau see "*A Monograph on Confidentiality and Privacy in the U.S. Census*" (Gatewood, 2001).

A considerable amount of research is specifically conducted on how to best implement methods to protect the confidentiality of respondents while also providing for the disclosure of useful results from analyses run on internal (restricted-access) census data. The following provides a brief overview on the gradual evolution of disclosure avoidance practices at the U.S. Census Bureau for the past 15 years with special emphasis on two applicable methods for this study: traditional privacy methods and mathematically provable private methods. Traditional methods include, but are not limited to, rounding, cell suppression, and top-coding. Mathematically provable private methods include, but are not limited to, differential privacy. If differentially private noise injection is applied, traditional privacy methods such as rounding are not needed (McKenna and Haubach, 2019). Zayatz (2005) provides a description of the state of affairs for disclosure avoidance in 2005 in which both rounding and the injection of noise are referenced as being practiced at the U.S. Census Bureau for both the Decennial Census and the ACS. Zayatz et al. (2010) provide an updated outline for both the 2010 Decennial Census and 5-year ACS disclosure avoidance practices. Rounding is mentioned in this publication, but noise injection is not. Ramanayake and Zayatz (2010) discuss cell suppression, data swapping (U.S. Census Bureau, 2010a), and the utilization of synthetic data as methods for limiting the risk of disclosure in Census Bureau data products, with no mention of either rounding or noise injection. Both noise injection and rounding appear in Lauger et al. (2014) regarding both microdata and magnitude data. Abowd and McKinney (2016) focus entirely on the injection of noise to ensure the protection of confidentiality and privacy through the use of graph theory and graph-based statistics, while Abowd and Schmutte (2017) again describe noise injection techniques with specific mention of the Laplace distribution and Laplace mechanism for injecting noise into data while maintaining a useful data product. Both the rounding and injection of noise into data values seem to have lost favor and then become more popular again within the last five years, with the adoption noise injection seemingly becoming far more widely researched and implemented than simple rounding.

Currently, the rounding of all requested output is enforced in adherence to U.S.C. Title 13. The following is an excerpt detailing the methods for disclosure rounding from Abowd (2018, p. 1) found in "*Summary Statistics/Model Based Output*":

> All estimates in released outputs must be rounded to four significant digits, base 10. Four significant digits is defined as $x.yyyE \pm nnn$, where $1 \leq x <= 9$, $0 \leq yyy \leq 999$, and nnn is the exponent. The numbers $1,234,000$, 1234, 1.234 and 0.0001234 all have four significant digits.

Table 1: "Special Rounding Rule for Entity Counts and Related Integers" from Abowd (2018), where N are entity counts. The entity count rounding applied to the location-aggregated MAF data in this paper, described in subsection 3.1, is highlighted in light gray.

N observations	Report for output
< 15	$N < 15$
$15 < N < 99$	round to nearest 10
$100 < N < 999$	round to nearest 50
$1000 < N < 9999$	round to nearest 100
$10\,000 < N < 99\,999$	round to nearest 500
$100\,000 < N < 999\,999$	round to nearest 1000
$1\,000\,000 < N$	round to four significant digits

Special rounding rules are applied specifically to entity counts, shown in Table 1. The row from Table 1 highlighted in gray is the rule applied to the population data used in this paper. This is due to the number of location-aggregated housing units being somewhere between 91,250 and 91,749, which is rounded to 91,500 and will be detailed later in Table 6. The data itself and the process for preparing and validating the data will be described in the following section.

A disclosure avoidance (DA) review process is required before any output (e.g. statistical outputs, notes, programs, qualitative summaries, and data products) may be released outside of the U.S. Census Bureau or a Federal Statistical Research Data Center (FSRDC). The Disclosure Review Board (DRB) sets the standard disclosure avoidance rules and policies required for review. The DRB operates under the guidance of the Data Stewardship Executive Policy Committee (DSEP). For more information see U.S. Census Bureau (2019b). A Disclosure Avoidance Officer (DAO) can use their delegated authority to directly approve the public release of some types of requested output without a formal DRB review. For any reason, if a DAO is unable to approve the release of output then they will bring it before the DRB for a decision. This is especially true for any requests that do not meet standard rules or are otherwise complicated.

In 2018, DSEP ruled that sub-state estimates must include a DRB-approved noise injection method unless they have a specific DSEP exemption or if they receive a DRB exception. Within this context, sub-state refers to a geographic area that is less populous than the least populous state. An analysis on Leon County, FL is sub-state because the population of this county was less than the population of the least populous state in 2010. The least populous state in 2010 was Wyoming with a publicly-published population of 563,626.

When requested output includes sub-state estimates, output often requires noise injection. Output using noise injection requires approval by the DRB, the Chief Scientist, and sometimes an external reviewer. Differential privacy is an accepted mathematical concept developed and used for detecting and preventing risks to privacy, and signifying that data are provably private (Dwork and Roth, 2014). It is a burgeoning research topic in computer science and is implemented for determining

the extent to which certain data can be used to predict other data. Concerning the implementation of differential privacy for the release of statistics and data by the U.S. Census Bureau, there are currently two dominant schools of thought. Ruggles et al. (2019) assert that the implementation of differential privacy is not necessary or useful. Moreover, Ruggles et al. (2019) predicts that it will actually be harmful to future social and economic research initiatives with unforeseen repercussions, and that through the implementation of differential privacy the U.S. Census Bureau is violating its statutory obligations. Conversely, Abowd et al. (2017) describe the implementation of differential privacy within the context of a statistical agency (e.g., the U.S. Census Bureau) as a "match made in heaven" that will only get better due to differential privacy being an emerging field. These conflicting opinions are seen by many as the unavoidable growing pains of introducing a novel process into a matured system. Concerning this line of research, the results pertaining to the MAF data detailed later in this paper were some of the first specifically geographically-related restricted-access census microdata that were deemed in need of differential privacy through noise-injection techniques. For a general primer on differential privacy see Wood et al. (2018). Hsu et al. (2014) detail a method specifically for determining a proper level of epsilon (ε), which will be addressed in the next section, and Gaboardi et al. (2013) discuss linear dependent types.

This section reviewed and described abstract concepts of confidentiality and privacy that are enforced by the U.S. Census Bureau as part of its responsibility to uphold Title 13, Section 5 of the U.S. Code. The next section details the concrete methods in which two types of census microdata, the MAF and the Decennial Census, were prepared and validated within a secure Federal Statistical Research Data Center (FSRDC) prior to being cleared for results disclosure by the DRB.

3 Methods

3.1 Processing Confidential Microdata

The output of analysis using confidential or restricted data must follow the appropriate disclosure avoidance techniques before being released. Therefore, it is paramount to rigorously document the processing of the data in order to clearly understand the meaning and limitations of the output. To this end, a specific process was developed for this project that details the parameters and procedure of preparing the data starting with the initial raw microdata, cleansing the data while accounting for population and missing values, validating counts against disparate sources, and preparing the results obtained from the cleaned data for disclosure review and release. The parameters and 9-step procedure utilized for preparing and validating the census microdata used in this paper are described in Process 1[5].

Three sets of census microdata are used as inputs for the approach described in the **Parameters** section of Process 1: the 2017 Master Address File (MAF), 2010 Decennial Census Population file (DECP), and 2010 Decennial Census Housing file (DECH). The 2017 MAF was used because it was most current form of the MAF. As mentioned in section 1, MAF records are continually updated and

[5]This process of noise injection and results disclosure is specific to FSRDC Project Number **at1866**. Other projects may follow similar steps, but the procedure described in Process 1 is specific to this research project.

Parameters

- 2010 Decennial Census Population file (DECP)
- 2010 Decennial Census Housing file (DECH)
- 2017 Master Address File (MAF)

Procedure

Step 1: Prepare MAF Data (2017 data available by state)

- Subset Florida file to Leon County
- Subset to records contained in the 2010 Decennial Census
- Drop MAF records with no associated population

Step 2: Prepare DECP and DECH Data (2010 data available by state)

- Subset Florida file to Leon County

Step 3: Validate Decennial Census Data

- For each DECP and DECH, compare population total to MAF population total

Step 4: Combine MAF records by location

- Assign MAF records Census GEOIDs and a longitude-latitude ID
- Group MAF records by unique longitude-latitude ID and aggregate population
- Drop aggregated MAF records with no associated longitude-latitude

Step 5: Re-validate population counts

- Repeat Step 3

Step 6: Analyses

- See Table 6 and Table 7
- See Figure 14a and Figure 14b

Step 7: Inject aggregated MAF population with Laplace noise

- See (1), (2), and (3), and demonstrations in Fig. 1 and Fig. 15 of Appendix C

Step 8: Perform disclosure rounding

- See Table 1, specifically the row highlighted in gray

Step 9: Receiver Operating Characteristic calculations

- See demonstration in Fig. 2

Process 1: Cleansing and validating confidential census microdata followed by noise injection and rounding for a disclosure request.

never removed. Therefore, additional clarifying information, such as longitude and latitude, may be available in the years following a census, potentially providing a more complete base of records. All results obtained from Process 1 are based on the MAF, while the DECP and DECH are used for validating population counts through the procedural steps.

Beginning with step 1 in the **Procedure** section of Process 1, the 2017 MAF for the entire state of Florida was obtained. This file was subset down to only the records (individual housing units) contained within Leon County and further subset to records that were included in the 2010 Decennial Census. Next, all records without an associated population were dropped. The exact figures for MAF record totals and records dropped were not disclosed and reporting is prohibited. Following the initial scrubbing of the MAF, the DECP and DECH are prepared in step 2. The internal 2010 DECP and DECH, like the 2017 MAF, are available by state. Therefore, they must be subset down to Leon County, also like the 2017 MAF. After the completion of the initial cleansing, the population totals of the MAF, DECP, and DECH are compared for validation in step 3. This is an essential act for maintaining both the integrity of the cleansing process and the data itself.

Step 4 of Process 1 represents the transition in the workflow from a sanitization phase to a generative phase. All MAF records have the potential to be associated with a longitude and a latitude, but some MAF records may share an identical location, like all the housing units in an apartment building. Therefore, a longitude-latitude identification marker is generated for each MAF record. Following this generation, all unique MAF records were combined and their populations were aggregated. At this point all MAF records without an associated longitude and latitude are dropped before another data integrity validation is performed in step 5.

Step 6 represents the phase in which the actual core research presented later in this paper is performed. The data products from step 6 include Table 6, Table 7, Figure 14a, and Figure 14b, which are described in section 4. Following analysis, steps 7 and 8 of Process 1 acknowledge that any requests to disclose results must be reviewed by the DRB. All results must follow specific rules that may include special rounding procedures, seen in Table 1, as well as noise injection and implementing differential privacy, explained in section 2.3. In step 8 noise from a Laplace distribution is injected into aggregated households that have already been allocated to network segments. The process for allocating the aggregated households to the network, $hh2n$, will be explained fully in section 4.

The true set of aggregated population counts is processed to produce the $hh2n$ points. The ten noise-infused versions of this dataset are created using a Laplace distribution, (1). The scale value (the magnitude of change) of λ is defined by the ten values of ε, seen in (2) and (3), respectively. The variable value for each record and population mean of variable values for all records are given by x and μ, respectively, in (1).

$$f(x \mid \mu, \lambda) = \frac{1}{2\lambda} \exp\left(-\frac{|x-\mu|}{\lambda}\right) \tag{1}$$

$$\lambda = \frac{1}{\varepsilon} \tag{2}$$

$$\varepsilon = [1, 2, \ldots, 9, 25] \tag{3}$$

For a complete description of the Laplace distribution see Forbes et al. (2011, Ch. 26) and Stockute and Johnson (2013). At the time of this writing, the requirement for noise injection at the substate level (see section 2.3) was relatively new.

Following an evaluation from the Disclosure Review Board, and approval from the U.S. Census Bureau's Chief Scientist and an external reviewer, the dataset generated from one value of ε was granted permission to be disclosed. However, the specific value of ε may not be disclosed. The evaluation was based on a review of the Receiver Operating Characteristic (ROC) curves using the Root Mean Square Error (RMSE), (4), of population-weighted network segments and the Total Bias, (5), of segment to segment network shortest path distances.

$$\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(d_i - p_i\right)^2} = RMSE \tag{4}$$

$$\sum_{i=1} d_i - p_i = TotalBias \tag{5}$$

In (4) and (5), each element of the set, represented as n, is indexed over $i = 1$ where d are Laplace noise-infused values and p are the true data values. These two metrics were used to measure accuracy. The generation and evaluation of the ROC curve is performed in step 9 of Process 1. An example of an ROC curve generated from a toy dataset using the RMSE as the accuracy metric will be provided next, along with plots demonstrating the shape of the Laplace noise that is infused into the data.

Consider the "true" values of a toy dataset in column *Data* of Table 2 along with the Laplace–infused noise values. The dataset values are consecutive integers that range from 0 to 9,999 with Laplace noise injected into each value of *Data* ten times: $\varepsilon = 1$ through $\varepsilon = 9$ and $\varepsilon = 25$. The process for generating Laplace noise is found in (1), (2), and (3). As seen in Fig. 1, the $\varepsilon = 25$ noise displays the characteristic symmetry and leptokurtic shape of the Laplace distribution (Kotz et al., 2001, Ch. 2). Fig. 15 in Appendix C further demonstrates this shape with $\varepsilon = [1, 2, \ldots, 9]$.

Table 2: A toy dataset with a range of $[0, \ldots, 9999]$. Random Laplace noise, seen in (1), is injected into each value in *Data* with a λ of $1/\varepsilon$, seen in (2), for $\varepsilon = [1, 2, \ldots, 9, 25]$, seen in (3). The distribution of $\varepsilon = 25$ is visualized in Fig. 1 and Fig. 15 in Appendix C shows the distribution for each level of $\varepsilon = [1, 2, \ldots, 9]$. The Root Mean Square Error is taken from each column of ε values and used to create Fig. 2.

Data	$\varepsilon = 1$	$\varepsilon = 2$	$\cdots$	$\varepsilon = 9$	$\varepsilon = 25$
0	−0.028 302	0.482 406	$\cdots$	0.077 212	0.015 212
1	1.908 514	0.869 087	$\cdots$	0.972 619	1.053 390
2	3.106 563	3.021 132	$\cdots$	1.723 599	1.969 976
⋮	⋮	⋮	$\ddots$	⋮	⋮
9997	9994.704 563	9996.583 768	$\cdots$	9996.890 133	9997.090 043
9998	9998.013 793	9997.854 503	$\cdots$	9997.988 147	9998.005 177
9999	9999.215 906	9998.999 773	$\cdots$	9999.055 002	9999.183 830

The Root Mean Square Error (RMSE), seen in (4), is used as an example accuracy metric and calculated for each column of ε. The resulting RMSE values are plotted against each other to produce a Receiver Operating Characteristic curve, as shown in Fig. 2. The x–axis of Fig. 2 is each value of ε 1 through 9 and the y–axis is $RMSE_{\varepsilon_{25}} - RMSE_{\varepsilon_i}$ for each i in 1 through 9. The area under an ROC curve is considered "bad" in terms of a higher likelihood of data at that level of ε being able to predict the true data values. Therefore, in this toy dataset, the Laplace noise injected with $\varepsilon = 1$ has the absolute least likelihood of being used to compromise privacy. In other words, as the level of ε that is utilized to calculate λ decreases, the risk of privacy loss decreases and vice versa (Abowd, 2019a,c). Abowd (2019b) argues that the benefits of using this differential privacy technique greatly outweighs the perceived drawbacks by providing privacy guarantees that are public and explainable. For a complete, general primer on ROC curves and their interpretation see Gneiting and Vogel (2018).

This section described the specific process used to prepare and validate internal census microdata. The process for data preparation and validation is performed prior to manipulating the dataset to ensure conformity with the U.S. Census Bureau's strict rules that protect each respondent's privacy and guarantee respondent confidentiality with advanced differential privacy techniques in accordance with Title 13 of the U.S. Code. Having knowledge of these techniques is not necessary, but helpful, for understanding the dataset. Likewise, having some background familiarity of the study area's context (e.g., history and regional information) is helpful. This information is provided in the following subsection.

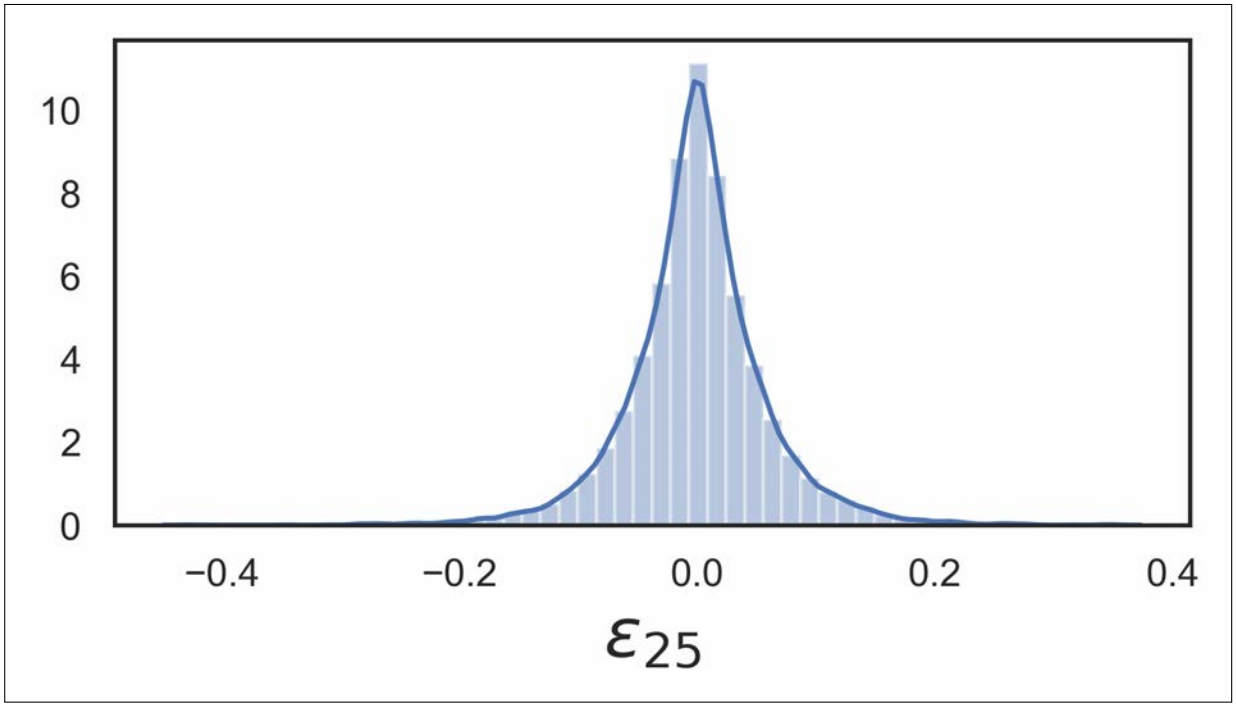


Figure 1: A demonstration of the characteristic leptokurtic symmetry of the Laplace distribution in the noise injected into the toy data from Table 2 for $\varepsilon = 25$. The x–axis represents the magnitude of noise and the y–axis represents probability.

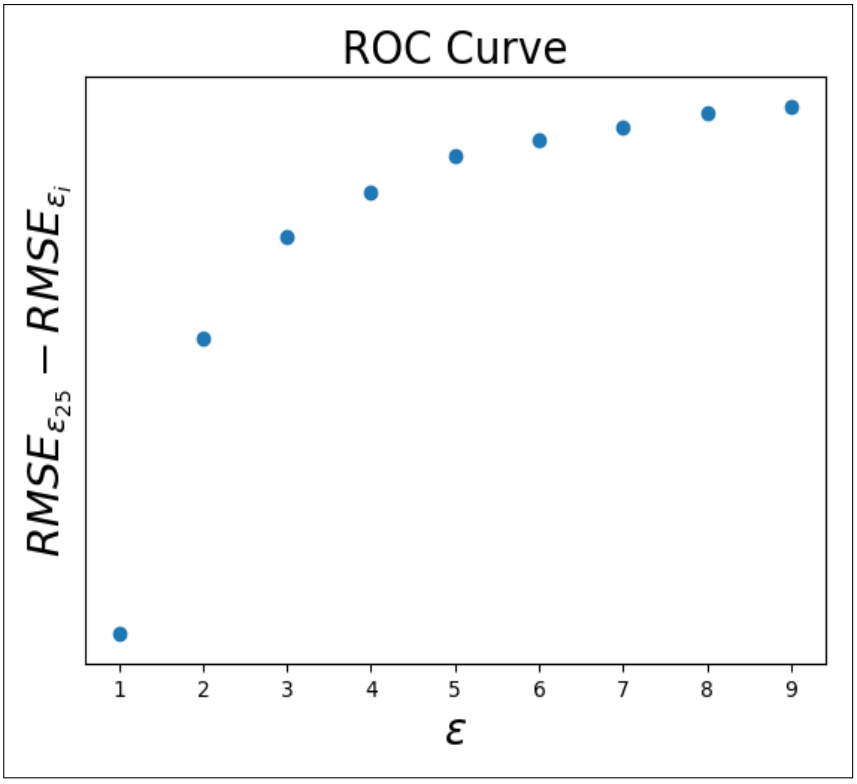


Figure 2: An example Receiver Operating Characteristic (ROC) curve generated for the toy dataset shown in Table 2. The purpose of an ROC curve is to determine the likelihood of a particular dataset being able to predict the values of another dataset.

3.2 Context of the Study Area

In this section the context of the full study area will be briefly described with a combination of satellite imagery, traditional maps, and text. Understanding the regional and local context of a study area for an empirical testing of methods provides a helpful supplement for validating this study. The ability to contribute personal knowledge regarding the terrain, road network, and individual neighborhoods within Leon County, and specifically Tallahassee, extends the usefulness of the resultant knowledge. All information regarding population, boundaries, and areas was current in 2010.

Tallahassee is situated in the southeast United States in the Panhandle of North Florida. It was chosen as the state capital of Florida in 1824, due to its central location between the two largest cites in the state at the time: Pensacola and St. Augustine (Leon County Division of Tourism, 2019), as seen in Fig. 3. While Pensacola and Jacksonville (St. Augustine is far less important than it was 200 years ago) are still major urban areas in Florida they are less prominent than the four nearest large metropolitan statistical areas (MSAs) that are between 4 and 6 hours drive away: Orlando, FL, Tampa, FL, New Orleans, LA, and Atlanta, GA.

Fig. 4 displays Leon County, FL in which Tallahassee is the county seat within the Florida Panhandle. Leon County straddles the Georgia border and is near the coast of the Gulf of Mexico. Specifically, Leon County lies within the largely-rural Big Bend (Coast) region and is generally considered to be part of the Deep South, unlike much of Central and South Florida. Although there is no formal delineation of the region, it normally refers to the curvature of the coast from the Apalachicola Bay to Cedar Key, and the coastal and immediately inland counties.

Tallahassee is the primary city in the Big Bend Coast region and is also the core of the Tallahassee Metropolitan Statistical Area, which includes Leon County, Wakulla County to the south, Gadsden County to the west, and Jefferson County to the east.

As mentioned earlier, large areas within Leon County are rural, as seen in Fig. 5. The northeastern portion of the Apalachicola National Forest occupies the southwest of Leon County, along with the Lake Talquin State Forest in the west, and the Tall Timbers research institute in the north. Though Tallahassee is the only city in Leon County, Woodville is a census-designated place and there are numerous unincorporated communities such as Bradfordville, Capitola, Chaires, Centerville, Fort Braden, Iamonia, Meridian, and Miccosukee.

Fig. 6 depicts the Tallahassee city limits. The Tallahassee International Airport and Leon County Fairgrounds are clearly delineated in the extreme southwest and southeast city limits, respectively. The central urban area of the city is bounded by Interstate-10 in the north, which then becomes more suburban and rural. The northeastern most portion of incorporated Tallahassee is colloquially known as Killearn, though it actually is comprised of many smaller subdivisions.

The downtown corridor is clearly visible in Fig. 7 with Florida State University to the immediate west and Florida Agricultural and Mechanical University, an HBCU (historically black college or university), to the south. The community of Frenchtown, a historically African American community, lies to the northwest of downtown Tallahassee.

According to the publicly-published 2010 Decennial Census, Leon County had a total area of 1,817.5 km^2 of which 1,727.08 km^2 were land (95.02%) and 90.55 km^2 were water (4.98%). When comparing urban and rural land in Leon County, 81.17% (1,475.24 km^2) of the total land area was

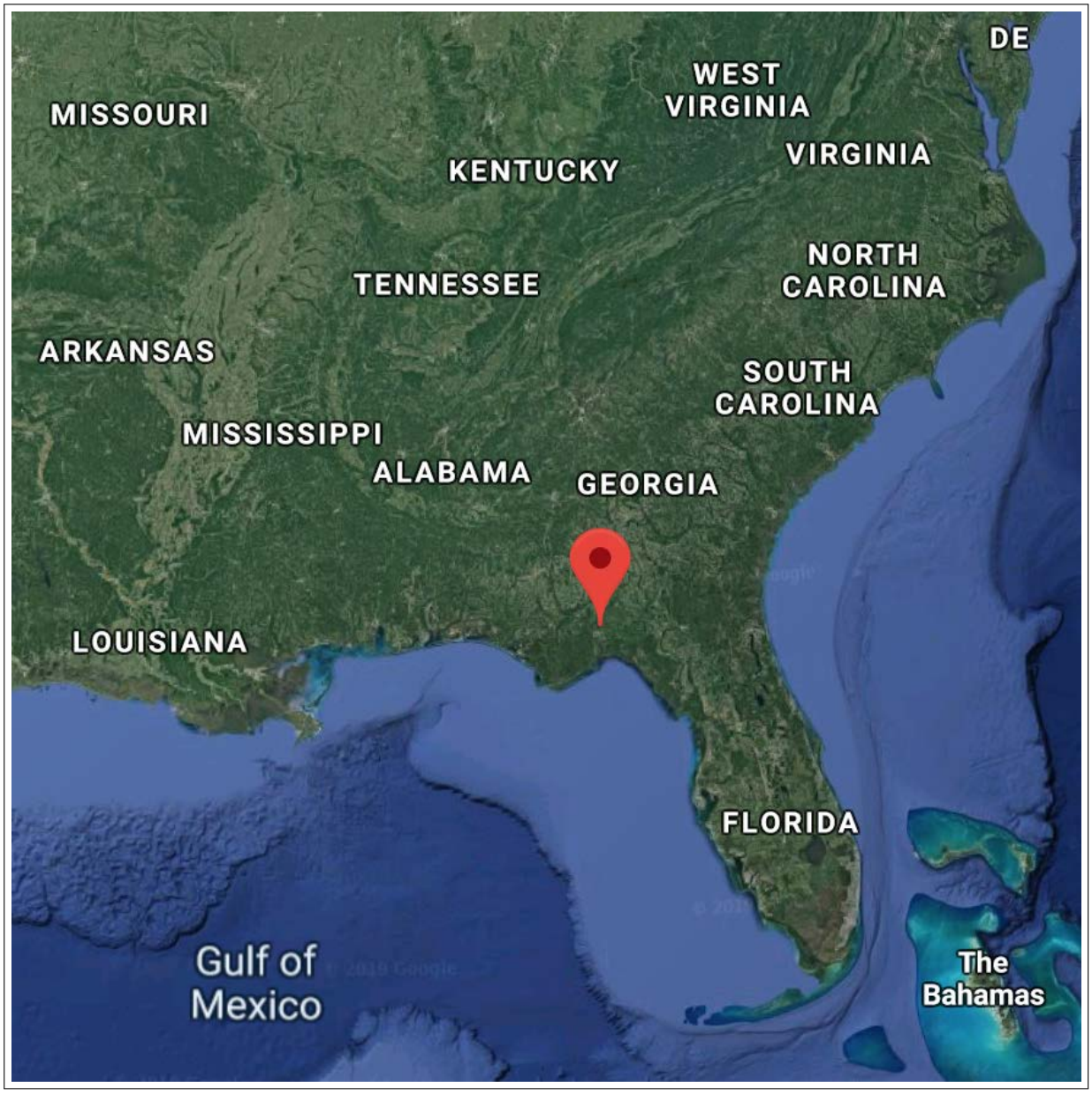


Figure 3: Satellite imagery of the southeast United States, including the Gulf Coast and the southern portion of the Eastern Seaboard. Tallahassee is shown with a red pin in the largely-rural northwestern portion of Florida known as the Florida Panhandle. Image from Google (2019e).

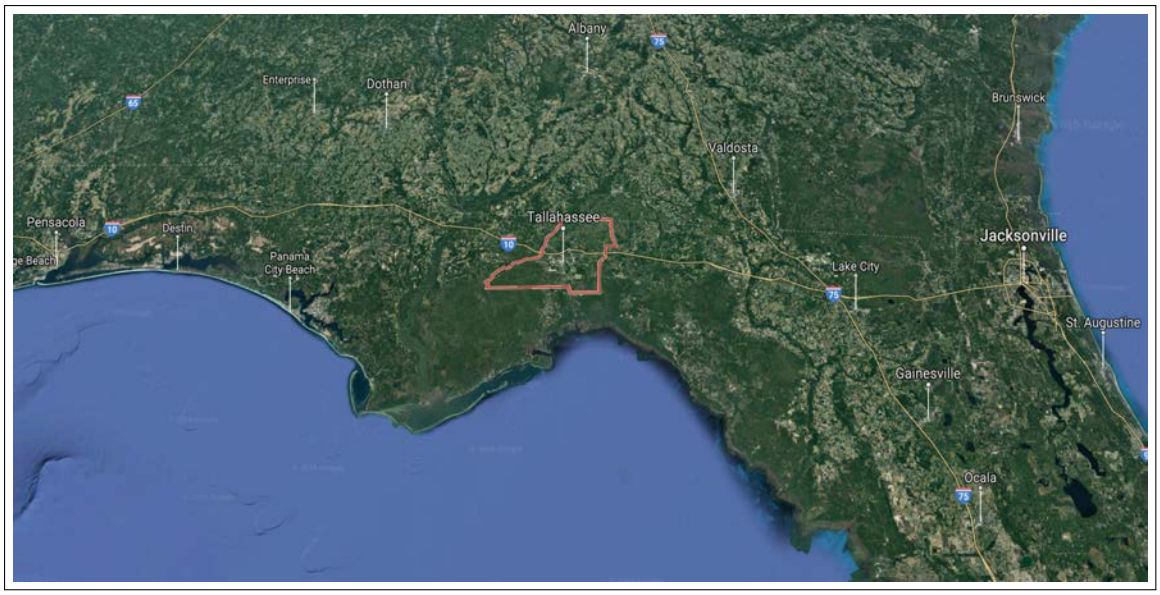


Figure 4: Satellite imagery demonstrating the position of Leon County, FL (outlined in red) within the Big Bend (Coast) region of North Florida. Image from Google (2019a).

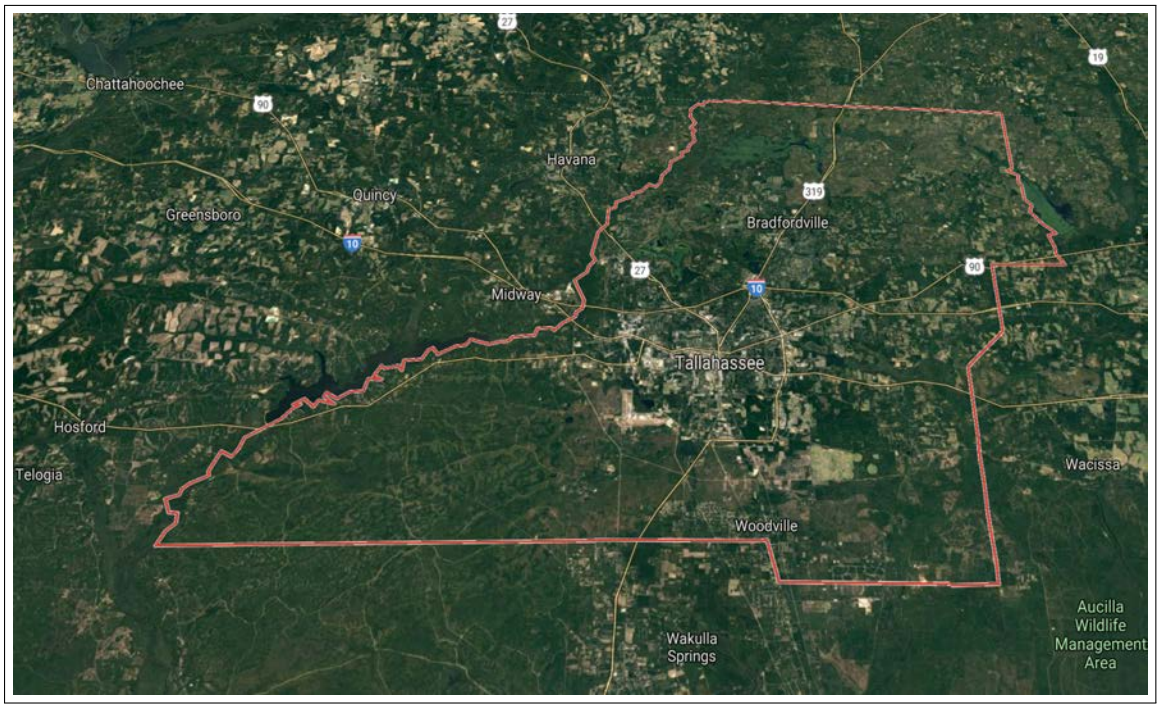


Figure 5: Satellite imagery of Leon County, FL (outlined in red) and outlying rural areas. Image from Google (2019d).

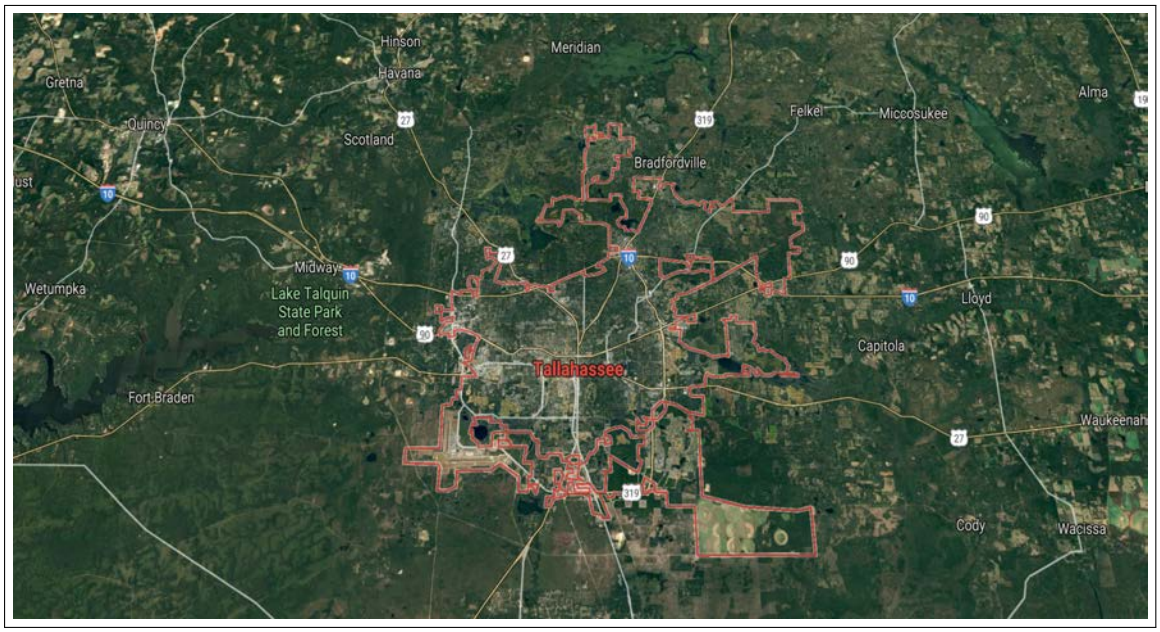


Figure 6: Satellite imagery of the city limits of Tallahassee, FL (outlined in red) and outlying rural areas. Image from Google (2019c).

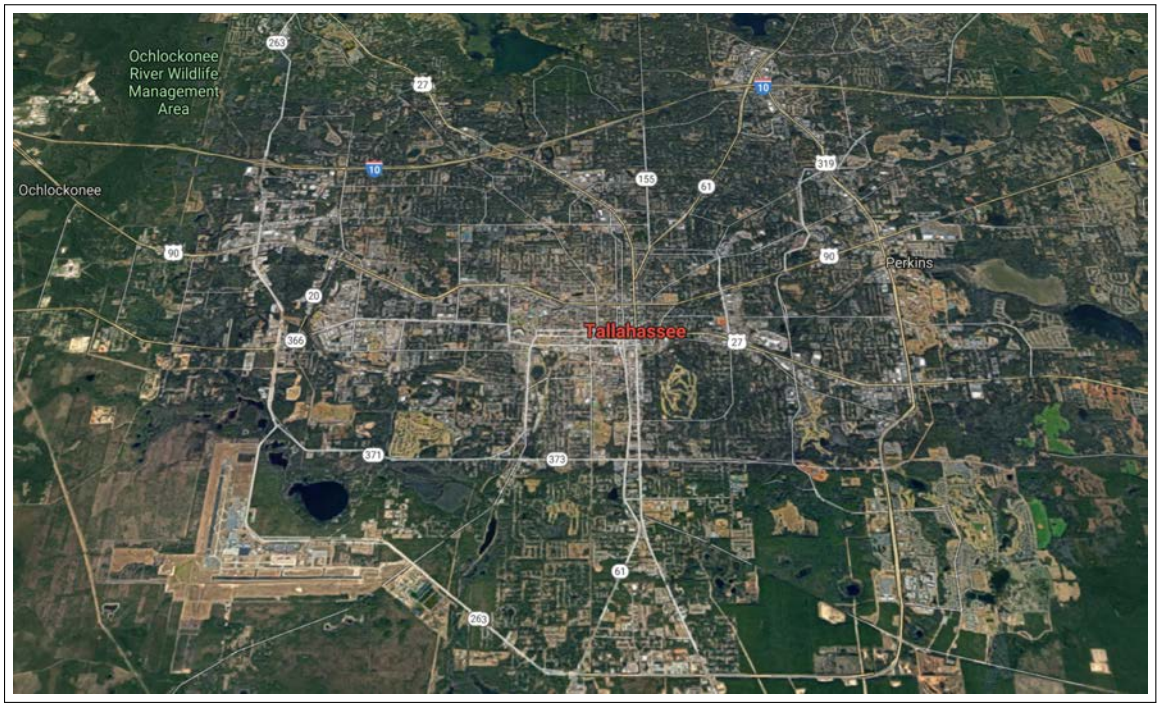


Figure 7: Satellite imagery of the central urban area of Tallahassee, FL. Image from Google (2019b).

rural and 18.83% (342.26 km^2) was urban (U.S. Census Bureau, 2012). As of 2010 there was a recorded population of 275,487 accounting for a population density of 159.50 people per km^2, where only land area is considered, demonstrated by (6).

$$\frac{Population_{Total}}{Area_{Land}} = Population_{Density} \tag{6}$$

Also, there were 124,136 households accounting for 71.88 households per km^2, which follows the same form as (6), seen in (7).

$$\frac{Households_{Total}}{Area_{Land}} = Households_{Density} \tag{7}$$

In 2010 the city of Tallahassee had a total area of 267.94 km^2, 259.63 km^2 (96.9%) of which was land and 8.31 km^2 (3.1%) water. A population of 181,376 and 74,815 households were recorded in 2010 accounting for a population density of 698.57 people per km^2 and a household density of 288.15 households per km^2, using (6) and (7), respectively (U.S. Census Bureau, 2019a). As is expected, much higher population and household densities are observed within the city of Tallahassee than in Leon County as a whole.

Describing Leon County in terms of purely census geographies there are 68 tracts and 177 block groups as seen in Fig. 8, where *POLY* represents the number of geographies (polygons). Tracts are shown with thick, transparent green borders and block groups are shown with thin blue borders, with the area occupied by the city of Tallahassee displayed with a golden transparency.

When comparing the imagery in Fig. 5 to the traditional map in Fig. 8, and taking the previous descriptive statistics into consideration, it is clear to see that rural tracts and block groups are larger than their urban counterparts, as is expected due to tracts and block groups being based on population and containing blocks as part of the standard census hierarchy (U.S. Census Bureau, 2018a). Furthermore, the smaller census tracts and blocks groups are generally contained within the city limits of Tallahassee and their larger counterparts without.

Moving down the census geography hierarchy from tracts and block groups, Fig. 9 shows census blocks where *POLY* represents the number of blocks (polygons). There were 6,198 blocks in Leon County in 2010, 3,634 of which were populated and 2,564 were not populated. Consistent with higher-level aggregations in Fig. 8, blocks which are larger tend to be in outlying, rural areas and tend to be associated with no population. A cluster of small, unpopulated blocks are observed in the central, downtown area of Tallahassee where the capitol building is located along with numerous businesses, as well as law and lobbying firms.

In contrast to census blocks, the population landscape appears somewhat less clustered when the spatial unit is changed to property parcels, as seen in Fig. 10. As described in Gaboardi (2019, Ch. 2), this dataset is the product of an advanced dasymetric preprocessing approach introduced by Strode et al. (2018) with two input polygon datasets, the 2010 U.S. census blocks and 2011(2010-certified) Florida Department of Revenue property parcels, which are overlaid to estimate individual

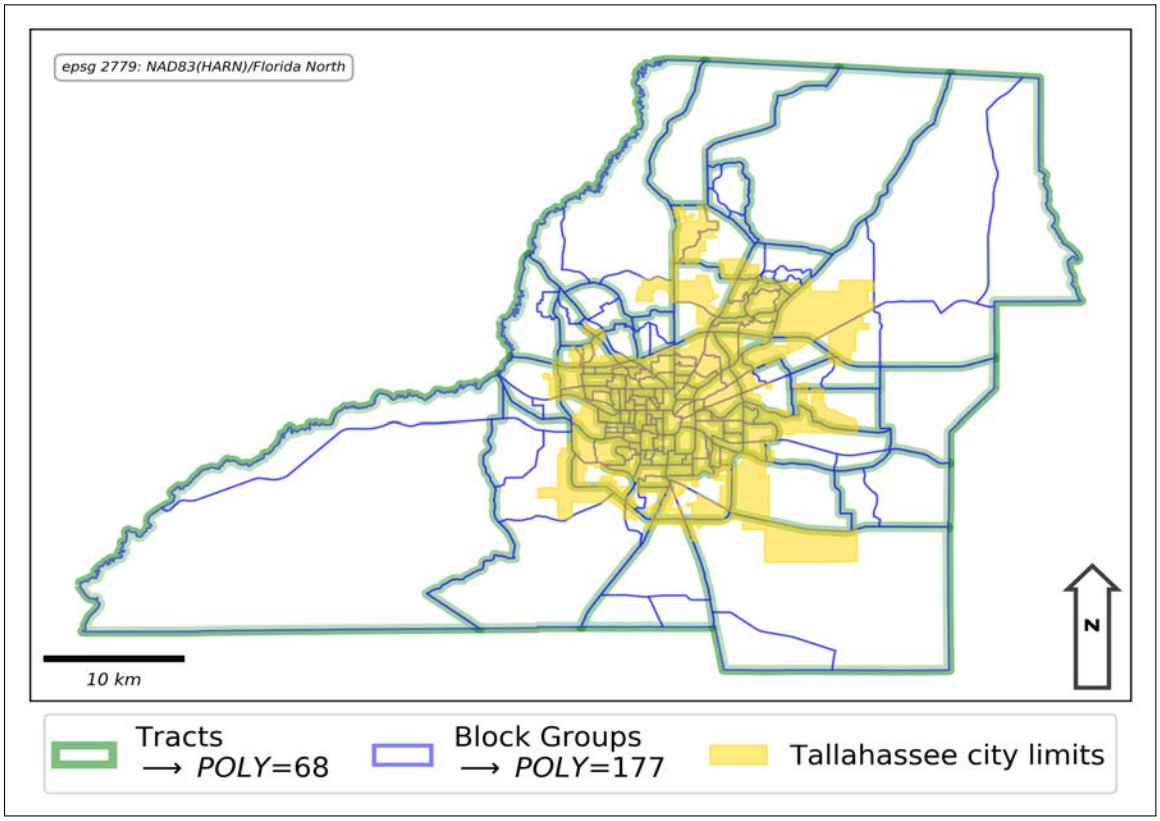


Figure 8: 2010 Decennial Census tracts and block groups within Leon County, FL. *POLY* represents the total number of polygons in each set of census geographies. The Tallahassee city limits are highlighted in gold.

property parcel populations. In 2010, there were 107,683 total parcels of which 87,877 (81.607%) were estimated to be reliably populated and 17,950 (16.669%) were estimated to be reliably unpopulated, shown in Fig. 10 with transparent and red polygons, respectively. Three property parcels, accounting for 0.002% of the total and symbolized in solid black polygons, were excluded from the dataset due to their centroids not being contained within any census geography associated with Leon County. The three property parcels had a combined estimated population of 0.0, and therefore, their exclusion was determined to have a neutral effect. The fourth category of features in Fig. 10 is displayed with a green and white hatch and represents property parcels that have an ambiguous population estimate. There are 1,853 of these parcels in the data, which account for 1.720% of the total. The concept of an ambiguous population estimate will be described in the next section and an example will be given in Fig. 12.

According to the description above and the legend of Fig. 10, several discrepancies become clear, all of which seem related and involve unclear population estimates. The two most obvious inconsistencies are the total population and total estimated population not being equivalent (275,487 vs. 275,465) and a negative estimated population sum in unpopulated parcels, -2.0899×10^{-11}. As was noted previously, this dataset was produced through a dasymetric process and dasymetrically-derived data have the potential to greatly enhance spatial resolution, but may also propagate uncertainty through unchecked inaccuracies (Nagle et al., 2014).

As was shown and described in the immediately preceding paragraphs, there appears to be an issue with the process that was utilized to generate the population-weighted property parcel dataset.

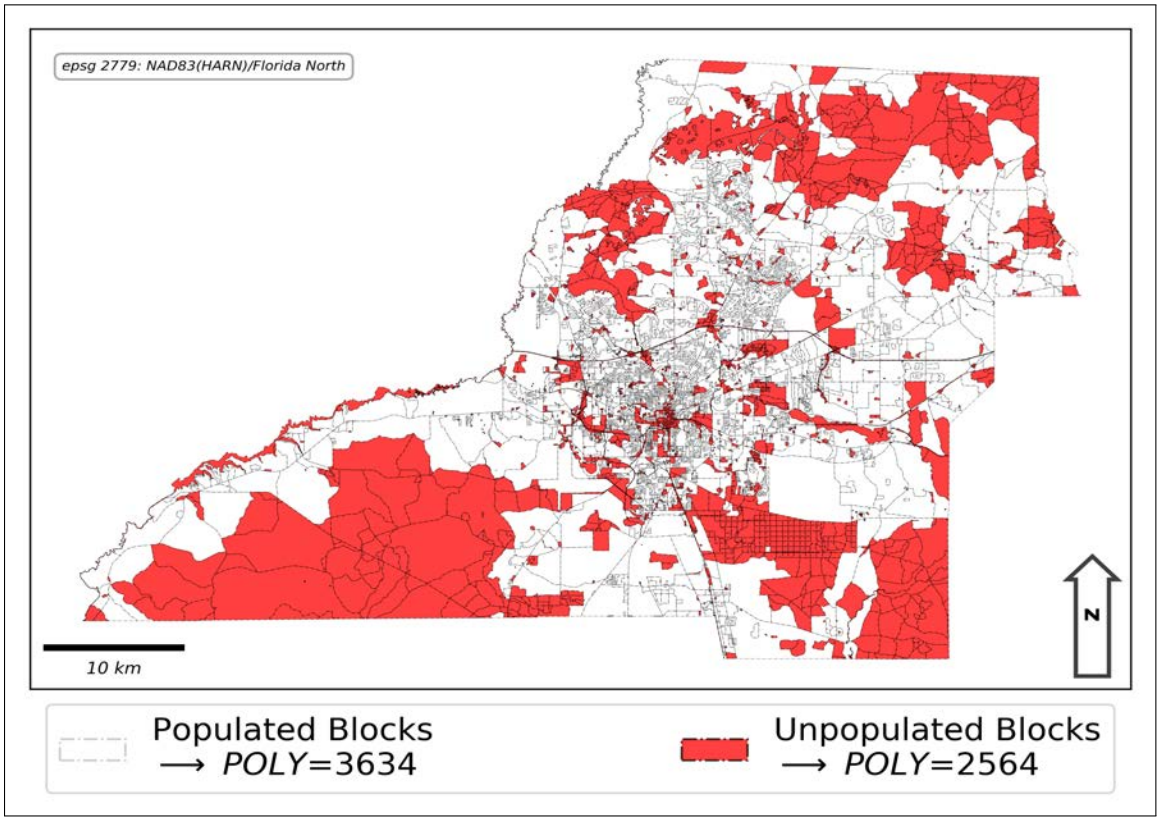


Figure 9: 2010 Decennial Census blocks within Leon County, FL. Blocks with no associated population are highlighted in red. *POLY* represents the total number of polygons in each set of census geographies.

The following subsection describes and critiques the process by which Strode et al. (2018) produce the estimated population weights for property parcels, which are the base for the $wp2n$ method.

3.3 A Critique of a Dasymetric Method

Strode et al. (2018) describe an innovative extension to the Cadastral-based Expert Daysmetric System (CEDS) proposed by Maantay et al. (2007). CEDS is an established methodology by which property records and census geographies are dasymetrically-combined for the purpose of obtaining population distributions at a higher spatial resolution. The proposed extension is a preprocessing method that is focused on the accurate assessment of population in group living quarters. The U.S. Census Bureau defines group quarters as being either institutional or non-institutional. Institutional group quarters are correctional facilities, nursing homes, and mental hospitals. Non-Institutional group quarters are college dormitories, military barracks, group homes, missions, and shelters (U.S. Census Bureau, 2018b). Strode et al. (2018) identify Leon County, FL as having a group quarters population twice that of the national average, though this figure appears to be from 2017, not 2010.

The preprocessing extension is performed in four steps beginning with the creation of a parity table (step 1). This table aligns the cadastral data and census data by property-use codes and types. Next, an intersect operation is performed whereby the attributes of census geographies are assigned to property parcels (step 2) then the population of group living quarters is attributed to

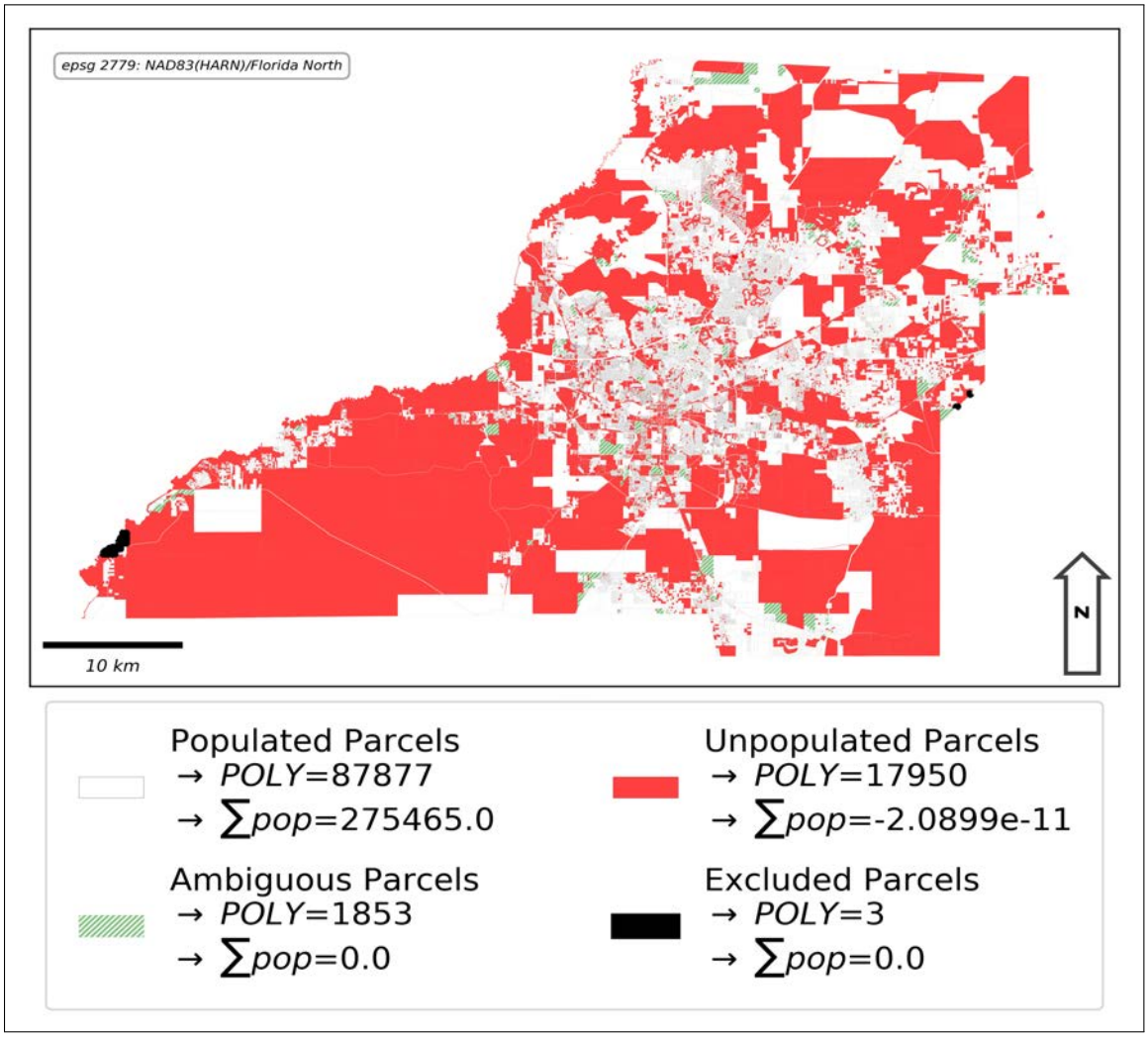


Figure 10: 2011(2010-certified) property parcels within Leon County, FL. Parcels with an absolute estimated population of zero are highlighted in red, while parcels with an ambiguous estimated population of very near zero are hatched with green bars (see Figure 12). Three parcels are excluded.

the appropriate property parcels based on the table created in step 1. Finally, the CEDS workflow is executed based on the initial formulation found in Maantay et al. (2007). This series of steps is summarized in Process 2.

Fig. 11 represents the base network in Leon County and the dataset that is allocated to the network when performing the $wp2n$ method. There were a total of 18,787 simplified segments and 15,065 vertices in the Leon County network resulting from Process 1.1 described in Gaboardi (2019, Ch. 1). The $wp2n$ dataset represents the results of the preprocessing technique while excluding the centroids of reliably unpopulated property parcels (17,950). The remaining parcel centroids, 89,730, are categorized as either weighted or unweighted, symbolized with purple and red squares, respectively. Parcel centroids are considered weighted when associated with a reliable population, otherwise they are considered unweighted. This concept was visualized in Fig. 10 with green and white hatched polygons. Comparing Fig. 10 and Fig. 11, clustering appears to be more visible when representing parcels with their geometric centroids. Also, the clustering of "ambiguously" (un)weighted parcels within the central area on Tallahassee becomes more visible in Fig. 11.

Parameters

- Cadastral parcels
- Census geographies

Procedure

Step 1: Create parcel-to-geography equivalency table
Step 2: Intersect census geographies with property parcels
Step 3: Apportion population based on Step 1
Step 4: Perform CEDS

Process 2: A preprocessing extension to the Cadastral-based Expert Daysmetric System (CEDS) described by Strode et al. (2018). A higher-quality population distribution assigned to group living quarters is central to this extension.

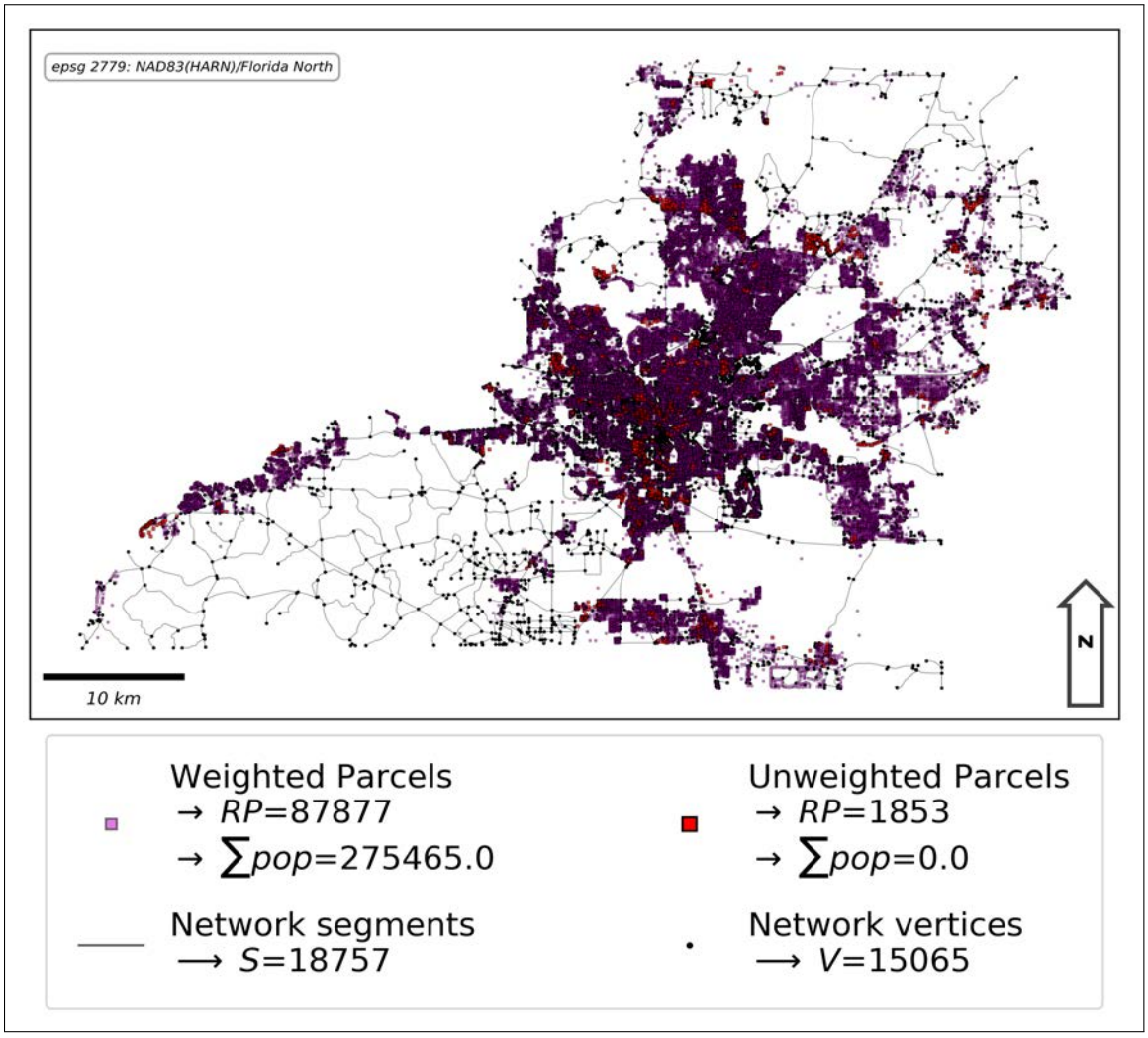


Figure 11: 2011(2010-certified) population-weighted property parcel centroids ($wp2n$) within Leon County, FL and the intersecting network structure, $N = (S, V)$. The totals for elements in each of the sets are given by RP ($wp2n$ representative points), S (network segments), and V (network vertices). Parcel centroids that were determined to have an ambiguous population are shown with red squares (see Fig. 10).

Before moving forward, two issues must be further discussed here: 1) the discrepancy between the total recorded population in the 2010 Decennial Census and the estimated population as calculated by this preprocessed CEDS approach, and 2) the concept of "ambiguously" (un)weighted parcels. First, the issue of total population discrepancy, which was raised in the previous section, deals with an exiguous underestimate. This underestimate is only a population of 22, representing less than $8 \times 10^{-5}\%$ of the total. Therefore, the result is still considered reasonable, but the underlying issue causing the underestimate must be investigated. The campus of Florida State University was used in the empirical example given in Strode et al. (2018) where only one out of a known population of 6,543 was not recorded when using the preprocessing technique. Also, the error at the campus scale actually accounts for a greater percentage than that at the county scale, approximately $1.53 \times 10^{-4}\%$. Furthermore, this error was due to mismatching population counts between the census blocks population and the group quarter population. Therefore, there appears to be a lower likelihood that similar errors may be manifested at the county level where there are fewer instances of group living quarters as compared to central Tallahassee.

Next, the issue of "ambiguously" (un)weighted parcels must be addressed. This issue appears to be caused by the CEDS procedure in which some property parcels are assigned a population that is virtually zero, but still greater than zero. A specific example of this is Parcel #1118208020000 with an estimated population of 0.00000000000 as represented and displayed in QGIS (QGIS Development Team, 2016), but was selected as greater than zero programmatically. The method implemented for slicing out parcel centroids with no population is demonstrated in Fig. 12, specifically lines 4–12. Interestingly, this parcel is Roselawn Cemetery, located in the Waverly Hills/Piedmont-Live Oak neighborhoods, where there should be absolutely no (living) population. As noted in the previous section, there are 1,853 of these unreliably unweighted property parcels.

Therefore, future work for the enhancement of the dataset generation proposed by Strode et al. (2018) may include improvements to both the CEDS procedure and a threshold for minimum population. Regarding the actual CEDS procedure, there is potential for incorporating the penalized maximum entropy dasymetric model (P-MEDM) proposed by Nagle et al. (2014). The P-MEDM, not to be confused with the p-median problem that are featured prominently in Gaboardi (2019, Ch. 4), is a daysmetric methodology that transfers error through the model and produces documented uncertainty. In terms of preventing the recording of incredibly minute population values, a small, positively-signed minimum threshold may be enforced to prevent ambiguous values from being recorded. There are two potential solutions to this issue. The more basic solution simply discards population values that are less than a very small user-defined threshold. This threshold is defined in lines 15–20 of Fig. 12 as the smallest representable float. A more thoughtful solution is the utilization of the multiplicative inverse of the total population as the threshold. For example, $1/\sum pop$ where $\sum pop$ is the total recorded population in the study area. In the case of the Leon County, FL in 2010 this equates to $1/275487 = 3.629\,93 \times 10^{-6}$, which is significantly more identifiable as not being zero than the values observed currently. With the addition of a minimum threshold, a very small number larger than zero may be used in place of zero to remove unweighted property parcels, as demonstrated in lines 22–28 of Fig. 12. As a further note, the concept of zero-weighted population representations are important for the analysis in Gaboardi (2019, Ch. 4) (optimal facility

```
import geopandas
import numpy

# read in parcel data as geodataframe
dataframe = geopandas.GeoDataFrame('path/to/data.shp')

# estimated population column
est_pop = 'SUM_EST_PO'
population = dataframe[est_pop]

# ----------------------------- current implementation
# slice out parcels with population greater than 0.0
dataframe = dataframe[population > 0.0]

# ---------------- potential future implementation (1)
# slice out parcels with population greater than
# the smallest positive floating point representable
# on a specific machine
SMALL = numpy.finfo(float).eps
dataframe = dataframe[population > SMALL]

# ---------------- potential future implementation (2)
# slice out parcels with population greater than
# the multiplicative inverse of the study area's
# summed population
sum_pop = population.sum()
SMALL = 1. / float(sum_pop)
dataframe = dataframe[population > SMALL]
```

Figure 12: An example of the current implementation in Python for slicing out property parcels with no population from a dataframe. Although syntactically unambiguous, incredibly small values (< 0.00000000000) may still be considered > 0.0, seemingly due to limitations in precision. One future implementation may utilize a very small epsilon, represented here as the variable SMALL defined as the smallest possible representable float, rather than true zero. Another implementation may utilize the population's multiplicative inverse as the baseline.

location modeling) for two reasons: (1) models are computationally more complex as the dataset size increases; and (2) the inclusion of clients with no associated population has the potential to skew the results of location-allocation models (Batta, 1989), even those in which population is not considered as a model parameter, the location set covering problem and p-center problem, for example. These reasons are further discussed in section 2.5 and 3.1 of Gaboardi (2019, Ch. 4).

This section described and critiqued the procedures by which the datasets utilized in this study were generated and the datasets themselves. Although the procedures and data were examined critically, they were deemed to be of a high quality. Scrutinizing workflows and products are vital to producing robust datasets and results, though it is important to understand that there will always be limitations. The results obtained from network allocation using the datasets examined in this section will now be given equal scrutiny in the following section.

4 Validation of Network Allocation Methods

Validation of the network allocation methods introduced in section 2.3 of Gaboardi (2019, Ch. 2) are performed in a similar manner as in 2.4 section of Gaboardi (2019, Ch. 2), but at the full county level. Descriptive statistics of the population associated with network segments for all methods of network allocation will be provided and discussed. This will be done first with publicly-available data using the $wp2n$ generated dataset as a benchmark. Next, a review of the analysis from data within the secure RDC environment will be provided. This data is actual household-level data from the 2017 Master Address File (MAF) that were present in the 2010 Decennial Census. Results from the public data run from within the RDC are also included with the restricted microdata scenarios for a comparison in respect to the data adjustments associated with the disclosure preparation process. This comparison is taken one step further by directly comparing the results obtained from the public pseudo-benchmark data ($wp2n$) to the true benchmark data created from the MAF using the *households to network* method, which will hereinafter be referred to as $hh2n$ to facilitate naming uniformity. Finally, Euclidean distance as a ratio of network distance will be investigated at the county level, as in section 2.4.1 of Gaboardi (2019, Ch. 2). However, where OD pairs in Fig. 2.14 and Fig. 2.15 from Gaboardi (2019, Ch. 2) are from population representations to population representations, the origins in this paper are fire stations.

Regarding the selection of the $pp2n$ parallel offset (po) input, two po parameters were used to generate the $pp2n$ points at the county level: 5 meters and 36 meters, as is used in Gaboardi (2019, Ch. 2). Although the average snapped distance to parcels at the county level is 38.98 meters, a po of 36 meters is used again at the county scale for uniformity with Gaboardi (2019, Ch. 2). This po is also used in Gaboardi (2019, Ch. 4).

4.1 Publicly-Published Data

Network segment population statistics based on the Leon County, FL road network using only publicly-available data run from outside the RDC's secure computing environment are reviewed

here. This is in contrast to the following subsection that uses both publicly-available data and restricted-access data, and is run from inside the RDC. Table 3 and Table 4 are reproductions of Table 2.3 and Table 2.4 in Gaboardi (2019, Ch. 2). Basic descriptive statistics are utilized to relate the population distribution along the network and how it varies according to both the spatial extent of the census geographies used and the method of network allocation. Table 3 provides an overview of measures that are not affected by unweighted network segments, whereas Table 4 shows measures that are affected by unweighted network segments.

Beginning with Table 3, in terms of validity, all methods for representing populations along the road network produce equivalent sums at 275,487, which was the official publicly-published population total for Leon County in the 2010 Decennial Census, with the lone exception being the $wp2n$ method. The $wp2n$ method produced a population estimate of 275,465, which was explained in subsection 3.2 and subsection 3.3. The number of representative points (RP), the percentage of network segments with no associated population (%0 pop), the total network population (sum), and the maximum associated population of any individual segment can be seen. Unsurprisingly, there are significantly fewer RP associated with the $pc2n$ method. This is, of course, a function of the spatial aggregations where tracts contain block groups and block groups contain blocks. As mentioned previously in subsection 3.2, in the 2010 Decennial Census there were 68 tracts, 177 block groups, and 6,198 blocks (3,634 populated) in Leon County. Therefore, there are equal numbers of RP for each level of aggregation. However, the number of RP for the more high-resolution population representations increases dramatically. The $va2n$ and $pp2n$ methods (5m and 36m) all produce respectively equivalent RP at the tract and block group levels, with a slight decrease at the (populated) block level. These numbers can be explained by the procedures for generating the datasets producing at most one point per network segment for the $va2n$ method and two points per segment for the $pp2n$ method. As the $wp2n$ is the most fine-grained resolution, it is expected that a larger number of RP will be observed (89,730). There are approximately 1/3 the number of $pp2n$ points as there are of $wp2n$ points, and there are approximately 1/2 the number of $va2n$ points as there are of $pp2n$ points. The relationship between RP and "unpopulated" segments appears to display a quasi-inverse nature. For example, the vast majority on network segments (more than 99%) are unpopulated when using the $pc2n$ method for census tracts. This shows that the $wp2n$ benchmark is picking up the clusters of population where property parcels are more dense, whereas the $va2n$ and $pp2n$ are more evenly distributing the population across the entirety of the network. In terms of the maximum observed segment population, the $va2n$ and $pp2n$ methods all produce roughly equivalent values, within approximately 200 at each spatial extent. The $pc2n$ produces a large maximum of 9,064 (3.3% of the total county) at the tract level then declines rapidly to 1,496, while the maximum for the $wp2n$ method is 2,388.

Table 4 shows descriptive statistics that may be affected by the presence of zero values: the minimum, mean, and standard deviation of population values for network segments. As such, Table 4 is divided into two sections respresenting the inclusion of zero values and the exclusion of zero values. In agreement with Table 2.4 in Gaboardi (2019, Ch. 2), when including unpopulated segments the minimum population value observed for all methods of network allocation is zero. This is because in each method of network allocation there is at least one segment with no associated population.

Table 3: Summary statistics for segment-based population weights including the generated representative point count (RP), percentage of segments with no associated population (%0 pop), cumulative population of segments (sum), and maximum segment weight (max). All statistics are calculated from publicly-published and available data.

Method	RP	%0 pop	sum	max
		Tract-derived		
$pc2n$	68	99.636	275 487.000	9064.000
$va2n$	18 657	0.107	275 487.000	700.625
$pp2n$ (5.0)	37 311	0.466	275 487.000	558.672
$pp2n$ (36.0)	37 296	9.776	275 487.000	459.904
		Block Group-derived		
$pc2n$	177	99.051	275 487.000	4552.000
$va2n$	18 657	0.107	275 487.000	568.168
$pp2n$ (5.0)	37 311	0.466	275 487.000	459.216
$pp2n$ (36.0)	37 296	9.776	275 487.000	376.836
		Block-derived		
$pc2n$	3634	82.553	275 487.000	1496.000
$va2n$	17 425	6.700	275 487.000	1049.435
$pp2n$ (5.0)	32 491	7.708	275 487.000	1260.957
$pp2n$ (36.0)	33 150	16.353	275 487.000	1073.891
		Benchmark		
$wp2n$	89 730	31.254	275 465.000	2388.000

Similarly, all methods have a mean segment population of 14.766, with the exception of the slight deviation seen in the $wp2n$ at 14.765. In terms of standard deviation the values observed appear to follow a similar pattern to that seen in the maximum values of Table 3. That is, the $pc2n$ observed at the block group-level is approximately half that of the tract-level and 2.5 times that of the block-level, the $va2n$ and $pp2n$ values are all roughly equal and the $wp2n$ is slightly larger.

When excluding segments with no associated population the most noticeably affected method of network allocation is the $pc2n$. For example, when zeros are masked from the dataset, the minimum population value increases from zero to 1,292, which changes the mean population value of the dataset from 14.766 to 4,051.279 and the standard deviation from 268.742 to 1,874.322. Similar patterns are seen with $pc2n$ points generated from block groups and blocks. However, the remainder of the population representation methods are affected significantly less by the masking of zero values with all values differing by less than eight from their respective unmasked values. An interesting observation can be seen when comparing the minimum values of the $pc2n$ method to all other methods. Minimums of 1,292 and 478 are observed the tract-level and block group-level, respectively, which are vastly larger than any other minimums, including $pc2n$ blocks. Though these two minimum segment population values are so much larger than the any others, this is an expected outcome because the base of a $pc2n$ point is simply the population-weighted geometric centroid of a single census geography and the smallest tract and block group in Leon County, FL in 2010 had populations of 1,292 and 478, respectively. A minimum of one is observed in the $pc2n$ scenario derived from blocks. Compared to all other methods the next largest minimum, observed at the tract level of $pp2n$ (5),

is only 4.5% of the one observed at the block level of *pc2n*. A global minimum of 1.0×10^{-15} is observed, which is known to cause issues in the recognition of population as demonstrated in Fig. 12 of subsection 3.3.

Table 4: Summary statistics (minimum, mean, and standard deviation) of segment population weight. The differences between including all network segments and masking segments with no associated population weight becomes clear. All statistics are calculated from publicly-published and available data.

Method	All segments			Populated segments		
	min	mean	std	min	mean	std
			Tract-derived			
pc2n	0.000	14.766	268.742	1292.000	4051.279	1874.322
va2n	0.000	14.766	28.908	5.456×10^{-5}	14.782	28.919
pp2n (5.0)	0.000	14.766	17.508	0.045	14.835	17.519
pp2n (36.0)	0.000	14.766	18.352	0.038	16.366	18.631
			Block Group-derived			
pc2n	0.000	14.766	167.706	478.000	1556.424	753.777
va2n	0.000	14.766	26.779	1.241×10^{-4}	14.782	26.789
pp2n (5.0)	0.000	14.766	17.818	0.020	14.835	17.831
pp2n (36.0)	0.000	14.766	18.887	0.020	16.366	19.214
			Block-derived			
pc2n	0.000	14.766	68.406	1.000	84.635	144.614
va2n	0.000	14.766	26.263	7.868×10^{-7}	15.826	26.879
pp2n (5.0)	0.000	14.766	23.958	0.002	15.999	24.540
pp2n (36.0)	0.000	14.766	25.963	9.843×10^{-4}	17.653	27.476
			Benchmark			
wp2n	0.000	14.765	45.134	1×10^{-15}	21.477	53.095

Table 5 shows the runtime comparisons for the *va2n* and *pp2n* methods, as are demonstrated in Table 2.2 and Table 2.5 from Gaboardi (2019, Ch. 2). With a 5-meter *po*, the *pp2n* method runs in 1.402%, 1.518%, and 22.171% of the time of the *va2n* method for census tracts, block groups, and blocks, respectively. The runtimes remain stable with a 36-meter *po* at 1.459%, 1.592%, and 22.575% under the same conditions stated in the previous sentence. Also, the percentage of time for calculating the line Voronoi diagram (LVD) within the *va2n* method is slightly less in general than observed in the empirical example shown in Table 2.5.

Two major observations can be made regarding runtimes in Table 5. First, when compared to the empirical example runtimes in Table 2.5, runtimes at the county level perform similarly. Proportionally similar runtimes at one census tract and a full county suggest promise in the scalability of the *pp2n* algorithm. Second, *pp2n* representations from blocks take approximately 13 times longer to generate than from other geographies (tracts and block groups). This is due to the population search in the *pp2n* algorithm, step 2d of Process 2.1 in section 2.3.1 from Gaboardi (2019, Ch. 2). However, the iterations of step 2 in Process 2.1 are not dependent on the results of other iterations, meaning that this phase of the algorithm can be considered *embarrassingly parallel* and is thus a strong candidate for the introduction of multiprocessing functionality. Multiprocessing would allow

step 2 of the *pp2n* algorithm to be split across all CPU processors, thus improving real runtimes on personal computers, and especially on HPC clusters.

Table 5: Comparative runtimes for the *va2n* and *pp2n* methods within Leon County, FL. Each method of population representation is generated from publicly-available 2010 Decennial Census tracts, blocks groups, and blocks, shown in Fig. 8. and Fig. 9. The percentage of total time for the line Voronoi diagram generation needed for the *va2n* method is given in the second row. Time is measured in minutes.

Aggregation Level	Tracts	Block Groups	Blocks
va2n total	475.411	437.072	415.520
va2n LVD %	84.323	90.022	94.369
pp2n (5.0)	6.666	6.633	92.125
pp2n (36.0)	6.937	6.960	93.803

The scenarios performed on public data as shown in Table 5 were run in the Florida State University Research Computing Center (FSU RCC, 2019). On the High Performance Computing (HPC) cluster, node hpc-i32-8 was used, which has a 2.2 GHz AMD Opteron(tm) Processor 6174, 131.8 GB of total available system memory, and 48 cores. If multiprocessing had been implemented on this cluster node a speedup for step 2 of the *pp2n* algorithm of roughly 48 times could be expected. However, in truth the speedup would be slightly less due to the computational burden of initializing and allocating the processes across all CPU cores.

4.2 Restricted-Access Microdata

The scenarios performed on restricted data (and public data) were run in the Atlanta Research Data Center (ARDC, 2019), one of the Federal Statistical Research Data Centers (FSRDC, 2019), on a virtual Red Hat Enterprise Linux (RHEL) 6.10-Santiago operating system (Red Hat, Inc., 2019) within the U.S. Census Bureau's Integrated Research Environment (IRE) (Fattaleh, 2017). Within the IRE, jobs were scheduled from a home (login) node and launched onto the HPC cluster using PBS Pro, a job scheduler and workload manager (Altair Engineering, Inc., 2019). Specific information for each processor and system memory was not recorded for restricted-access data scenarios, nor were the runtimes.

The results from the microdata discussed above in section 3.1 created through Process 1 will be discussed here and compared to the results from the previous section. The data found in this subsection was reviewed by the U.S. Census Bureau and poses no breach of confidentiality[6]. To reiterate, the analyses performed in this subsection are conducted using publicly-available data and restricted-access data run from within the RDC's secure computing environment. Table 6 is analogous to Table 3 and Table 7 is analogous to Table 4. The rounding rules shown in Table 1 and the Laplace noise-injection shown in Fig. 1 and Fig. 15 apply to Table 6 and Table 7. As is abundantly clear, all values calculated from the publicly-available data in Table 6 and Table 7 are very similar. However, when comparing the values of census geography-derived values to the

[6]The DRB approval number for the request pertaining to the data was CBDRB–FY19–387.

Table 6: Summary statistics for segment-based population weights including the generated representative point count (RP), percentage of segments with no associated population (%0 pop), cumulative population of segments (sum), and maximum segment weight (max). Statistics for the $hh2n$ method are calculated from internal census microdata. All others are calculated from publicly-published and available data, but within a secure environment.

Method	RP	%0 pop	sum	max
		Tract-derived		
$pc2n$	68	99.640	275 500.000	9064.000
$va2n$	18 650	0.107	275 500.000	700.600
$pp2n$ (5.0)	37 300	0.466	275 500.000	558.700
$pp2n$ (36.0)	37 290	9.773	275 500.000	459.900
		Block Group-derived		
$pc2n$	177	99.050	275 500.000	4552.000
$va2n$	18 650	0.107	275 500.000	568.200
$pp2n$ (5.0)	37 300	0.466	275 500.000	459.200
$pp2n$ (36.0)	37 290	9.773	275 500.000	376.800
		Block-derived		
$pc2n$	3634	82.560	275 500.000	1496.000
$va2n$	17 420	6.696	275 500.000	1049.000
$pp2n$ (5.0)	32 490	7.677	275 500.000	1261.000
$pp2n$ (36.0)	33 140	16.350	275 500.000	1074.000
		Benchmark		
$hh2n$	91 500	16.420	269 900.000	1333.000

$hh2h$ benchmark, the method that most accurately replicates the "truth" is $pp2n(36.0)$ derived from blocks. Considering the percentage of network segments with no population, and the minimally-populated and maximally-populated segment (excluding zero values), the statistics are virtually identical. Further, these results are replicated with 1/3 fewer representation points. Also, as appears to be the trend, the $pc2n$ method at the tract-level displays characteristics least similar to the $hh2n$ in terms of population distributions along the network.

Though the $pp2n(36.0)$ derived from blocks most closely mimics the $hh2h$ benchmark dataset derived from the MAF data, all methods for representing populations from publicly-available census geographies perform at least reasonably well. To illustrate this point somewhat more visually, consider (8) for calculating the percentage difference where ξ_{Public} and $\xi_{Restricted}$ are correspondent values for the same variable in the publicly-available data and restricted microdata, respectively, and $\Delta\%$ is the percentage of difference between the values.

$$\frac{|\xi_{Public} - \xi_{Restricted}|}{\left[\frac{\xi_{Public} + \xi_{Restricted}}{2}\right]} = \Delta\% \tag{8}$$

Table 8 and Table 9 demonstrate the percentage difference between each corresponding cell in Table 3 and Table 6, and again in Table 4 and Table 7. In Table 8 and Table 9 cells highlighted in green

Table 7: Summary statistics (minimum, mean, and standard deviation) of segment population weight. The differences between including all network segments and masking segments with no associated population weight becomes clear. Statistics for the *hh*2*n* method are calculated from internal census microdata. All others are calculated from publicly-published and available data, but within a secure environment.

Method	All segments			Populated segments		
	min	mean	std	min	mean	std
			Tract-derived			
*pc*2*n*	0	14.770	268.800	1292.000	4051.000	1874.000
*va*2*n*	0	14.770	28.910	5.456×10^{-5}	14.780	28.920
*pp*2*n* (5.0)	0	14.770	17.500	0.045	14.840	17.510
*pp*2*n* (36.0)	0	14.770	18.350	0.038	16.370	18.630
			Block Group-derived			
*pc*2*n*	0	14.770	167.700	478.000	1556.000	753.800
*va*2*n*	0	14.770	26.780	1.241×10^{-4}	14.780	26.790
*pp*2*n* (5.0)	0	14.770	17.800	0.020	14.840	17.810
*pp*2*n* (36.0)	0	14.770	18.890	0.020	16.370	19.220
			Block-derived			
*pc*2*n*	0	14.770	68.410	1.000	84.660	144.600
*va*2*n*	0	14.770	26.270	7.868×10^{-7}	15.830	26.880
*pp*2*n* (5.0)	0	14.770	23.960	0.002	16.000	24.530
*pp*2*n* (36.0)	0	14.770	25.970	9.843×10^{-4}	17.650	27.480
			Benchmark			
*hh*2*n*	0	14.470	37.310	1.493×10^{-4}	17.310	40.210

indicate no difference in generated data, black cells with white text indicate data generated outside the secure RDC environment expressed a larger observed value, while white cells with black text indicate that the data generated inside the secure RDC environment expressed a larger observed value.

As stated earlier and visualized in Table 8 and Table 9 the values calculated from the publicly-available data, both inside the RDC and outside, are virtually identical after considering the mandatory rounding to 4 significant digits of results, rounding of entity counts, and Laplace noise injection. In fact, the largest different in census geography-derived data observed in either Table 8 and Table 9 is the *pp*2*n*(5.0) produced from publicly-available data being 0.397% larger than the counterpart produced inside the RDC. Due to the equivalency shown in Table 8 and Table 9 it can be concluded that all of the methods themselves used here to produce population representations are sound and valid. Once the population representations based solely on census geographies are taken into account, the only substantive differences can be seen in the percentage differences between the two benchmark datasets: the *wp*2*n* and *hh*2*n*.

4.3 Weighted Parcels vs. Aggregated Households

The *wp*2*n* (*weighted parcels to network*) method was utilized as a measure to benchmark against the publicly-available census dataset, but how well does this pseudo-benchmark align with the true,

Table 8: Percentage difference (Δ%) between the summary statistics for segment-based population weights for Table 3 (public data) and Table 6 (restricted data and public data executed within a secure environment). Cells highlighted in green indicate equivalent values for the public and restricted data. Black cells with white font indicate the public value being greater than the restricted value. White cells with black font indicate the opposite.

	Δ%			
Method	*RP*	%0 pop	sum	max
		Tract-derived		
*pc*2*n*	0.000	0.004	0.005	0.000
*va*2*n*	0.038	0.002	0.005	0.004
*pp*2*n* (5.0)	0.029	0.019	0.005	0.005
*pp*2*n* (36.0)	0.016	0.036	0.005	0.001
		Block Group-derived		
*pc*2*n*	0.000	0.001	0.005	0.000
*va*2*n*	0.038	0.002	0.005	0.006
*pp*2*n* (5.0)	0.029	0.019	0.005	0.004
*pp*2*n* (36.0)	0.016	0.036	0.005	0.010
		Block-derived		
*pc*2*n*	0.000	0.008	0.005	0.000
*va*2*n*	0.029	0.058	0.005	0.041
*pp*2*n* (5.0)	0.003	0.397	0.005	0.003
*pp*2*n* (36.0)	0.030	0.019	0.005	0.010
		Benchmark		
*wp*2*n* ↔ *hh*2*n*	1.953	62.230	2.041	56.705

household-level MAF-derived data? According to the measures found in Table 8 and Table 9, the *wp*2*n* appears to nearly universally produce inflated estimates in population, with the sole underestimate being the minimally-populated segment in the network. However, upon closer inspection the true magnitude of these differences is relatively insignificant. For example, a 200.0% difference is observed for the minimally-populated segment with the *hh*2*n* displaying the larger value, but the values are microscopic fractions of population, 1.493×10^{-4} and 1×10^{-15}. The largest actual difference appears to be in the number of unpopulated segments, which would very likely have substantive implications when performing analyses based on the datasets. There are 31.254% unpopulated network segments in the *wp*2*n* dataset whereas the *hh*2*n* dataset has only 16.420%. This seems to suggest that the *wp*2*n* may be producing population along the network in clustered patterns that are not necessarily representative of reality. However, this has not been tested (and is purely speculation) and it is unlikely the results of such an analysis would ever be approved for disclosure due to privacy concerns regarding Title 13, Section 5 of the U.S. Code.

Considering all the results discussed thus far in this section, it can be concluded that the results obtained with the publicly-available, aggregated data, and population abstractions including the resultant datasets (*wp*2*n*) produced from the preprocessing CEDS method, are valid to a high degree of accuracy.

Table 9: Percentage difference (Δ%) between the summary statistics for segment-based population weights for Table 4 (public data) and Table 7 (restricted data and public data executed within a secure environment). Cells highlighted in green indicate equivalent values for the public and restricted data. Black cells with white font indicate the public value being greater than the restricted value. White cells with black font indicate the opposite.

	Δ%					
Method	All segments			Populated segments		
	min	mean	std	min	mean	std
			Tract-derived			
pc2n	0.000	0.028	0.021	0.000	0.007	0.017
va2n	0.000	0.028	0.008	0.004	0.012	0.003
pp2n (5.0)	0.000	0.028	0.044	0.010	0.033	0.053
pp2n (36.0)	0.000	0.028	0.012	0.010	0.025	0.006
			Block Group-derived			
pc2n	0.000	0.028	0.004	0.000	0.027	0.003
va2n	0.000	0.028	0.005	0.036	0.012	0.005
pp2n (5.0)	0.000	0.028	0.102	0.020	0.033	0.118
pp2n (36.0)	0.000	0.028	0.017	0.020	0.025	0.031
			Block-derived			
pc2n	0.000	0.028	0.005	0.000	0.030	0.010
va2n	0.000	0.028	0.027	0.003	0.024	0.003
pp2n (5.0)	0.000	0.028	0.006	0.011	0.006	0.041
pp2n (36.0)	0.000	0.028	0.026	0.003	0.015	0.015
			Benchmark			
$wp2n \leftrightarrow hh2n$	0.000	2.016	18.980	200.000	21.487	27.619

4.4 A Note on the Public vs. Restricted Discrepancy

There appear to have been a difference of ±5 *pp2n*(5.0) points generated on the same dataset with an identical code base within the secure RDC environment. Although the exact cause could not be determined, it is likely due to precision errors in measurement. This may be caused by a multitude of factors with the two main being different operating systems and slightly different versions of Python packages, and even more likely, a combination of the two. It is important to emphasize that although the exact number of representative points generated inside and outside the secure research environment may differ, the sum of population does not. This holds true for all abstract population representations generated from publicly-available data.

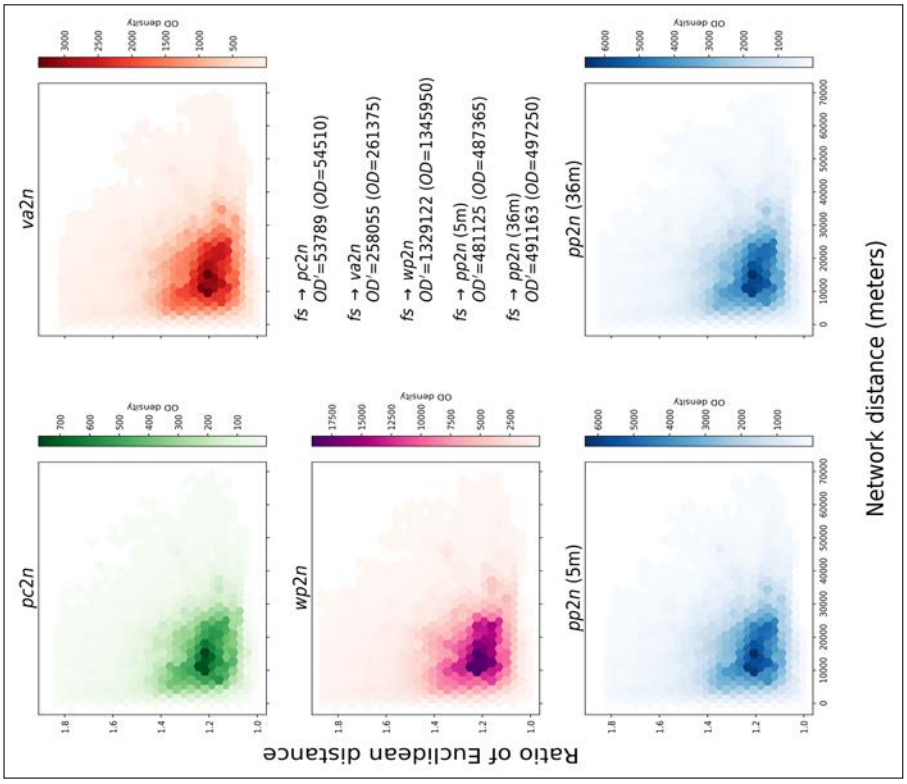
4.5 Euclidean Distance vs. Network Distance Revisited

The difference between network distance and Euclidean distance is shown to be non-trivial within the context of one census tract in Gaboardi (2019, Ch. 2), which is consistent with the results seen in Okabe and Sugihara (2012, Ch. 1). Unweighted and population-weighted distance are both used to describe these effects, seen in Fig. 2.14 and Fig. 2.15 from section 2.4.1 of Gaboardi (2019, Ch. 2), respectively. Here the analysis area is increased to the full county to determine whether the resulting distance ratios remain stable or tend to be more (or less) pronounced.

As stated previously, within the 25×25 hexbin plots found in the Gaboardi (2019, Ch. 2) all shortest path network cost matrices are calculated from population representations (origins) to population representations (destinations). In this paper the cost matrices used to determine the Euclidean distance as a ratio of network distance are calculated from fire stations (origins) to population representations (destinations). Also, like Gaboardi (2019, Ch. 2), all methods of allocating populations to the network are presented, but only at the census block level, and only the OD pairs within ± 3 standard deviations (σ) from the population mean (μ) are shown ($OD' \subset OD$). The context of fire stations within Leon County are be detailed in section 4.3.1 of Gaboardi (2019, Ch. 4) where fire stations represent service facilities for optimal facility location modeling. A total of 15 fire stations represent the origins of all OD' pairs within this subsection and also in Gaboardi (2019, Ch. 4). As in the comparison of Fig. 2.14 and Fig. 2.15 from Gaboardi (2019, Ch. 2) to the shape of Massachusetts, the shape of Georgia will be adduced and used as a mechanism to relate the 25×25 hexbin plots found in Fig. 13a, Fig. 13b, Fig. 14a, and Fig. 14b. To reiterate, these results do not actually concern Georgia. The data simply have a similar shape.

Beginning with Fig. 13a, there is a clear focal point of density occurring at between 10,000–20,000 meters of network distance accounting for a ratio of between 1.1 to 1.3 times that of the correspondant Euclidean distance. This focal point is approximately in the location of Albany in southwest Georgia and a large surrounding area with a diameter from Bainbridge to Cordele or thereabouts. This suggests that the shortest paths for the majority of all OD' pairs are only marginally longer over the road network than through Euclidean space. While the results here do show the network distance is generally longer than Euclidean distance, it appears to be much less pronounced when considering a full county than when only considering one census tract, as in Fig. 2.14 from Gaboardi (2019, Ch. 2). The likely cause of the difference in shape between Fig. 2.14 from Gaboardi (2019, Ch. 2) and Fig. 13a is that the OD pairs in Fig. 2.14 in Gaboardi (2019, Ch. 2) are from population representations to population representations, whereas the OD pairs in Fig. 13a are from fire stations to population representations. The functional purpose of fire stations is to be located within a close proximity to as many residences as possible, therefore, this result is unsurprising.

Next, population-weighted distance, shown in Fig. 13b, will be discussed. In Gaboardi (2019, Ch. 2), the image of a heavily-eroded coastline is invoked when describing the change in shape from Fig. 2.14 to Fig. 2.15. For Fig. 13a and Fig. 13b a politcal transformation will be used to describe the data shape. While all subplots in Fig. 13a are reminiscent of the state of Georgia in the southeastern U.S., Fig. 13b can be imagined as South Carolina, which is northeast of Georgia, encroaching on Georgia's border, pushing its own border to the southeast.



(b) Census block derived population-weighted network distance and the analogous ratio to population-weighted Euclidean distance in kilometers between fire stations and the midpoints of network segments. All zero-weighted (unpopulated) segment midpoints are excluded prior to extracting ± 3 standard deviations from the mean of each respective scenario's representation of population, shown by OD'. The scales for all axes are not uniform. 25×25 hexbins are used to demonstrate density, trends, and data shape.

(a) Network distance and the analogous ratio to Euclidean distance in meters between unique origin-destination ($ODʹ$) pairs of fire stations and population representations derived from census blocks. OD' are not the complete sets, but are subsets of OD (all pairs) within ± 3 standard deviations of each respective population mean (μ) for each scenario's representation of population. The scales for all axes are not uniform. 25×25 hexbins are used to demonstrate density, trends, and data shape.

Figure 13: Results from externally-processed data.

(a) Network distance and the analogous ratio to Euclidean distance in meters between unique origin-destination (OD') pairs of fire stations and population representations derived from census blocks. OD pairs for the $hh2n$ method are calculated from internal census microdata. All others are calculated from publicly-published and available data, but within a secure environment. OD' are not the complete sets, but are subsets of OD (all pairs) within ± 3 standard deviations of each respective population mean (μ) for each scenario's representation of population. The scales for all axes are not uniform. 25×25 hexbins are used to demonstrate density, trends, and data shape.

(b) Census block derived population-weighted network distance and the analogous ratio to population-weighted Euclidean distance in kilometers between fire stations and the midpoints of network segments. OD pairs for the $hh2n$ method are calculated from internal census microdata. All others are calculated from publicly-published and available data, but within a secure environment. All zero-weighted (unpopulated) segment midpoint are excluded prior to extracting ± 3 standard deviations from the mean of each respective scenario's representation of population, shown by OD'. The scales for all axes are not uniform. 25×25 hexbins are used to demonstrate density, trends, and data shape.

Figure 14: Results from internally-processed data.

Although the shape of the datasets and examples used to describe them differ, it appears that a similar underlying process results in the patterns seen in Fig. 2.15 from Gaboardi (2019, Ch. 2) and Fig. 13b, where the focal points are clustered in the east-southeast region of each "state." This seems to indicate that at both the scale of one census tract and a full county, the largest differences in network versus Euclidean population-weighted distance are observed between origin-destination pairs that are closer together (e.g. in urban areas). Also, when comparing the results of data produced outside the RDC and inside the RDC, the plots are virtually identical. This is shown in Fig. 14a and Fig. 14b.

5 Implications

There are two main implications that can be drawn from the review of the analyses in section 4. Also, an expected confirmation can be made regarding the publicly-available data from the 2010 Decennial Census: it appears to replicate the restricted-access microdata adequately. As for the implications, first, the dasymetric data ($wp2n$) is an acceptable proxy for actual census microdata. Second, since the aggregated, publicly-publish data and the dasymetrically-derived datasets appear to be sufficient substitutes for census microdata, a strong case can be made for them to be considered inherently valid (in terms of accurately representing true population distributions along networks).

The confirmation of expected results regarding the accuracy of publicly-available data suggests that previous work utilizing publicly-available population counts of census geographies does, in fact, bare a high degree of accuracy in representing the true nature of reality. However, this should not be misinterpreted as a blanket validation of all such research. It must be restated the that conclusions drawn from this research only come from tests within one county. Though a disclaimer must be made, these findings promote the concept of the validity of publicly-available census data for scientific research.

Next, the findings in this paper show that for Leon County, FL a high-quality proxy for the true, gold-standard MAF data can be generated through the use of the preprocessing technique and CEDS method proposed by Strode et al. (2018). Further, using this method for all counties in Florida is promising due to the uniform nature of the Florida Property Appraiser's data. However, each state in the U.S. implements different standards for maintaining property records. Therefore, generating a dataset similar to the $wp2n$ may prove to be problematic for another state. However, companies like Zillow that collect and aggregate property data may promote the feasibility of implementing the $wp2n$ method in other study areas (Zillow, 2019).

Finally, both points suggest that there may be a reduced demand for researchers to need access to restricted microdata, like the Master Address File, if comparable analyses can be made from publicly-available data. Obtaining the Special Sworn Status security clearance needed for access to the microdata is a time consuming, tedious process for both researchers and the U.S. Census Bureau. Also, having more researchers places a larger burden on the IRE's resources, and the FSRDC and Disclosure Avoidance staff. This may even imply the MAF is not as "restricted" as generally considered because it can be replicated with relative ease, and as noted in section 1, it

was originally based on mainly the USPS Delivery Sequence File. Considering these implications, ongoing research on this topic appears to be promising. However, to reiterate, though these are incredibly encouraging findings, they can only be said to apply to Leon County, FL for the data collected during the 2010 Decennial Census with any degree of certainty.

6 Conclusion

While researchers "have to learn to live with error, we should never ignore it" (Openshaw, 1989). Researchers must actively document error and uncertainty with the goal of validating spatial data while aggressively pursuing avenues in research that seek to diminish the effect of error and uncertainty. Although the utility of freely-available, publicly-published census data cannot be overstated, understanding the inherent uncertainty and ambiguity caused by the spatial aggregation of individuals into census geographies is equally important. Dealing with the limitations of uncertainty by investing considerable time and effort through a rigorous validation process is of the utmost consequence.

Through the research conducted here all generated population representations in this paper can be described as valid (to varying degrees) while each portrays the distribution of population slightly incongruously. So, considering their validity, how do the methods of network allocation described thus far, excluding the aggregated MAF-based $hh2n$ method, perform when being used as parameters in an applied setting? With ongoing research in Gaboardi (2019, Ch. 4), spatial optimization modeling are used to gain a better understanding of that question. Four fundamental facility location models are used to determine the varying outcomes of site selection when uncertain parameters, such population and location, are utilized for modeling purposes.

Appendix A List of Acronyms

The following is a list of acronyms frequently used throughout this paper.

ACS	American Community Survey
ARDC	Atlanta Research Data Center
CEDS	Cadastral-based Expert Daysmetric System
DECH	Decennial Census Housing file
DECP	Decennial Census Population file
FIPS	Federal Information Processing Series
FSRDC	Federal Statistical Research Data Center
GEOID	geographic identifier
GIS	Geographic Information Systems (also Science)
$hh2n$	households to network
HPC	high performance computing
LVD	line Voronoi diagram
MAF	Master Address File
MTFCC	MAF/TIGER Feature Class Code
OD	origin-destination, origin-to-destination
$pc2n$	polygon centroids to network
$pp2n$	populated polygons to networks
RDC	research data center
TIGER	Topologically Integrated Geographic Encoding and Referencing
TRP	transformed representation problem
$va2n$	vector area-to-network data conversion
$wp2n$	weighted parcels to network

Appendix B U.S.C. Title 13, Section 5: An Overview

The following excerpt summarizing the protections enacted by Title 13, Section 5 of the U.S. Code is taken directly from U.S. Census Bureau (2019c):

> Title 13 provides the following protections to individuals and businesses:
>
> 1. Private information is never published. It is against the law to disclose or publish any private information that identifies an individual or business such, including names, addresses (including GPS coordinates), Social Security Numbers, and telephone numbers.
> 2. The Census Bureau collects information to produce statistics. Personal information cannot be used against respondents by any government agency or court.
> 3. Census Bureau employees are sworn to protect confidentiality. People sworn to uphold Title 13 are legally required to maintain the confidentiality of your data. Every person with access to your data is sworn for life to protect your information and understands that the penalties for violating this law are applicable for a lifetime.
> 4. Violating the law is a serious federal crime. Anyone who violates this law will face severe penalties, including a federal prison sentence of up to five years, a fine of up to $250,000, or both.

Appendix C The Laplace Distribution

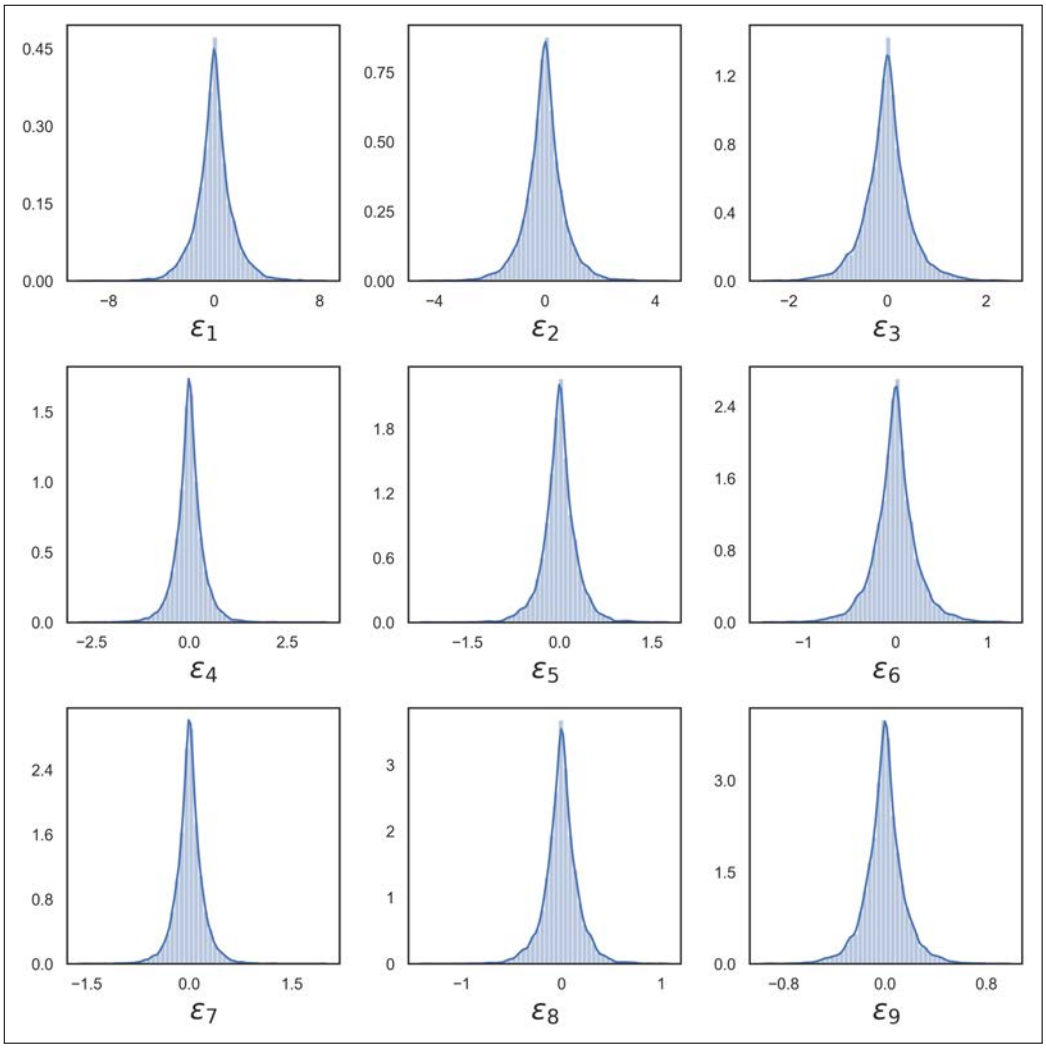


Figure 15: A demonstration of the characteristic leptokurtic symmetry of the Laplace distribution in the noise injected into the toy data from Table 2. Each subplot represents one level of $\varepsilon = [1, 2, \ldots, 9]$. The x–axes represent the magnitude of noise and the y–axes represent probability. The axes are not uniform.

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